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Technical Specification

3rd Generation Partnership Project; Technical Specification Group Radio Access Network; NR; User Equipment (UE) procedures in Idle mode and RRC Inactive state (Release 19)





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Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

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1 Scope

The present document specifies the Access Stratum (AS) part of the UE procedures in RRC_IDLE state (also called Idle mode) and RRC_INACTIVE state. The non-access stratum (NAS) part of Idle mode procedures and processes is specified in TS 23.122 [9].

The present document specifies the model for the functional division between the NAS and AS in a UE.

The present document applies to all UEs that support at least NR Radio Access, including multi-RAT UEs as described in 3GPP specifications, in the following cases:

- When the UE is camped on a NR cell;
- When the UE is searching for a cell to camp on;

NOTE: When the UE is camped on or searching for a cell to camp on belonging to other RATs, the UE behaviour is described in the specifications of the other RATs.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 38.300: "NR Overall Description; Stage 2".

[3] 3GPP TS 38.331: "NR; Radio Resource Control (RRC) - Protocol Specification".

[4] 3GPP TS 38.213: "NR; Physical layer procedures for control ".

[5] Void

[6] 3GPP TS 36.331: "E-UTRA; Radio Resource Control (RRC) - Protocol Specification".

[7] 3GPP TS 36.304: "E-UTRA; User Equipment (UE) procedures in RRC_IDLE state ".

[8] 3GPP TS 38.133: "NR; Requirements for Support of Radio Resource Management".

[9] 3GPP TS 23.122: "NAS functions related to Mobile Station (MS) in RRC_IDLE state".

[10] 3GPP TS 23.501: "System Architecture for the 5G System; Stage 2".

[11] 3GPP TS 38.215: "NR; Physical layer measurements".

[12] 3GPP TS 22.261: "Service requirements for the 5G system".

[13] 3GPP TS 24.890: "5G System – Phase 1; CT WG1 Aspects".

[14] 3GPP TS 24.501: "Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3".

[15] 3GPP TS 38.101-1: "NR; User Equipment (UE) radio transmission and reception; Part 1: Range 1 Standalone".

[16] 3GPP TS 23.287: "Architecture enhancements for 5G System (5GS) to support Vehicle-to-Everything (V2X) services".

[17] 3GPP TS 23.285: "Technical Specification Group Services and System Aspects; Architecture enhancements for V2X services".

[18] 3GPP TS 22.011: "Service accessibility".

[19] 3GPP TS 38.321: "NR; Medium Access Control (MAC); Protocol specification".

[20] 3GPP TS 26.517: "5G Multicast-Broadcast User Services; Protocols and Formats".

[21] 3GPP TS 23.247: "Architectural enhancements for 5G multicast-broadcast services; Stage 2".

[22] 3GPP TS 23.304: "Proximity based Services (ProSe) in 5G Systems (5GS)".

[23] 3GPP TS 23.003: "Numbering, addressing and identification".

[24] 3GPP TS 38.306: "User Equipment (UE) radio access capabilities".

[25] 3GPP TS 23.586: "Technical Specification Group Services and System Aspects; Architectural Enhancements to support Ranging based services and Sidelink Positioning".

[26] 3GPP TS 23.256: "Support of Uncrewed Aerial Systems (UAS) connectivity, identification and tracking; Stage 2".

3 Definitions, symbols and abbreviations

3.1 Definitions

For the purposes of the present document, the following terms and definitions apply:

2Rx XR UE: two antenna port XR UE as specified in TS 38.101-1 [15].

Acceptable Cell: A cell that satisfies certain conditions as specified in 4.5.

Allowed CAG list: A per-PLMN list of CAG Identifiers the UE is allowed to access (see TS 23.501 [10])**.**

Available PLMN(s): One or more PLMN(s) for which the UE has found at least one cell and read its PLMN identity(ies).

Available SNPN(s): One or more SNPN(s) for which the UE has found at least one cell and read its SNPN identity(ies).

Barred Cell: A cell a UE is not allowed to camp on.

CAG cell: A cell broadcasting at least one Closed Access Group Identifier.

Camped on a cell: UE has completed the cell selection/reselection process and has chosen a cell. The UE monitors system information and (in most cases) paging information.

Camped on any cell: UE is in idle mode and has completed the cell selection/reselection process and has chosen a cell irrespective of PLMN identity.

Child UE: A U2N Relay UE's next hop in downstream direction for serving a U2N Remote UE in U2N Relay communication. Child UE can be the U2N Remote UE or a U2N Relay UE.

Closed Access Group Identifier: Identifier of a CAG within a PLMN.

Commercial Mobile Alert System: Public Warning System that delivers *Warning Notifications* provided by *Warning Notification Providers* to CMAS capable UEs.

Downstream: direction toward child node or UE in IAB-topology or U2N Remote UE in U2N Relay communication.

Earth-fixed cell: An NTN cell fixed with respect to a certain geographic area on the earth all the time. It can be provisioned by beam(s) continuously covering the same geographical areas all the time (e.g., the case of GSO satellites).

Earth-moving cell: An NTN cell moving on the ground. It can be provisioned by beam(s) whose coverage area slides over the Earth surface (e.g., the case of NGSO satellites generating fixed or non-steerable beams).

eCall Only Mode: A UE configuration option that allows the UE to register at 5GC and register in IMS to perform only eCall Over IMS, and a non-emergency IMS call for test and/or terminal reconfiguration services.

EHPLMN: Any of the PLMN entries contained in the Equivalent HPLMN list TS 23.122 [9].

Equivalent PLMN list: List of PLMNs considered as equivalent by the UE for cell selection, cell reselection, and handover according to the information provided by the NAS.

Equivalent SNPN list: List of SNPNs considered as equivalent by the UE for cell selection, cell reselection, and handover according to the information provided by the NAS.

eRedCap UE: A UE with enhanced reduced capabilities as specified in clause 4.2.22 in TS 38.306 [24].

First U2N Relay UE: an Intermediate U2N Relay UE having both PC5 connection to a parent UE and PC5 connection to a U2N Remote UE for serving the U2N Remote UE in case of multi-hop L2 U2N Relay communication.

Home PLMN: A PLMN where the Mobile Country Code (MCC) and Mobile Network Code (MNC) of the PLMN identity are the same as the MCC and MNC of the IMSI.

HSDN cell: A cell that has higher priority than other cells for cell reselection for HSDN capable UE in a High-mobility state.

IAB-MT: As defined in TS 38.300 [2]. A mobile IAB-MT uses the same procedures and behaviours (including Cell Reservations and Access Restrictions) specified for the IAB-MT, unless explicitly stated otherwise.

Intermediate U2N Relay UE: a U2N Relay UE having both PC5 connection to a parent UE and PC5 connection to a child UE or a U2N Remote UE for serving the U2N Remote UE in case of multi-hop L2 U2N Relay communication.

Last U2N Relay UE: a U2N Relay UE having both Uu connection to the network and PC5 connection to a child UE for serving a U2N Remote UE in case of L2 U2N Relay communication.

Mobile-IAB cell: As defined in TS 38.300 [2].

Network Identifier: Identifier of an SNPN in combination with a PLMN ID (TS 23.501 [10]).

Non-Public Network: A network deployed for non-public use, as defined in TS 22.261 [12].

Non-terrestrial network: An NG-RAN consisting of gNBs, which provides non-terrestrial NR access to UEs by means of an NTN payload embarked on an airborne or space-borne NTN vehicle and an NTN Gateway.

NR sidelink communication: AS functionality enabling at least V2X Communication as defined in TS 23.287 [16] and/or A2X Communication as defined in TS 23.256 [26], and ProSe communication (including ProSe non-Relay, UE-to-Network Relay communication and, UE-to-UE Relay communication including UE-to-UE Relay communication with integrated discovery) as defined in TS 23.304 [22], between two or more nearby UEs, using NR technology but not traversing any network node.

NR sidelink discovery: AS functionality enabling ProSe non-Relay Discovery, ProSe UE-to-Network Relay discovery and ProSe UE-to-UE Relay discovery for Proximity based Services as defined in TS 23.304 [22] between two or more nearby UEs, using NR technology but not traversing any network node.

Paging adaptation: As defined in TS 38.300 [2].

Parent UE: A U2N Remote UE or U2N Relay UE's next hop U2N Relay UE in upstream direction for serving the U2N Remote UE in U2N Relay communication.

PC5 Relay RLC channel: an RLC channel between L2 U2N Remote UE and L2 U2N Relay UE, between L2 U2N Relay UEs (in case of multi-hop L2 U2N relay communication), or between L2 U2U Remote UE and L2 U2U Relay UE, which is used to transport packets over PC5 for L2 UE-to-Network/UE-to-UE Relay**.**

Process: A local action in the UE invoked by an RRC procedure or an RRC_IDLE or RRC_INACTIVE state procedure.

Quasi-Earth fixed cell: An NTN cell fixed with respect to a certain geographic area on the earth during a certain time duration. It can be provisioned by beam(s) covering one geographic area for a limited period and a different geographic area during another period (e.g., the case of NGSO satellites generating steerable beams).

Radio Access Technology: Type of technology used for radio access, for instance NR or E-UTRA.

Ranging/Sidelink Positioning: AS functionality enabling ranging-based services and sidelink positioning as defined in TS 23.586 [25].

RedCap UE: A UE with reduced capabilities as specified in clause 4.2.21 in TS 38.306 [24].

Registration Area: (NAS) registration area is an area in which the UE may roam without a need to perform location registration, which is a NAS procedure.

Registered PLMN: This is the PLMN on which certain Location Registration outcomes have occurred, as specified in TS 23.122 [9].

Registered SNPN: This is the SNPN on which certain Location Registration outcomes have occurred, as specified in TS 23.122 [9].

Reserved Cell: A cell on which camping is not allowed, except for particular UEs, if so indicated in the system information.

Selected PLMN: This is the PLMN that has been selected by the NAS, either manually or automatically.

Selected SNPN: This is the SNPN that has been selected by the NAS, either manually or automatically.

Serving cell: The cell on which the UE is camped.

Sidelink: UE to UE interface for V2X sidelink communication defined in TS 23.287[16].

SNPN Access Mode: Mode of operation wherein UE only selects SNPNs (as defined in TS 23.501 [10]).

SNPN identity: An identifier of an SNPN comprising of a PLMN ID and an NID combination.

Strongest cell: The cell on a particular frequency that is considered strongest according to the layer 1 cell search procedure (TS 38.213 [4], TS 38.215 [11]).

Suitable Cell: This is a cell on which a UE may camp. For NR cell, the criteria are defined in clause 4.5, for E-UTRA cell in TS 36.304 [7].

U2N Relay UE: a UE that provides functionality to support connectivity to the network for U2N Remote UE(s). Up to three L2 U2N Relay UEs (i.e. one Last U2N Relay and up to two Intermediate U2N Relays including one First U2N Relay) can be configured for serving a L2 U2N Remote UE in multi-hop L2 U2N Relay communication in this release.

U2N Remote UE: a UE that communicates with the network via one or more U2N Relay UEs on an indirect path.

U2U Relay UE: a UE that provides functionality to support connectivity for U2U Remote UE(s).

U2U Remote UE: a UE that communicates with other UE(s) via a U2U Relay UE.

UE-to-Network Relay communication: A mode of communication in which a UE communicates with the network through an indirect path involving only one U2N Relay UE for single-hop L2 U2N Relay communication or multiple L2 U2N Relay UEs for multi-hop L2 U2N Relay communication.

UE-to-Network Relay discovery: A mode of NR sidelink discovery in which a UE discovers other UEs for U2N Relay communication.

V2X sidelink communication: AS functionality enabling V2X Communication as defined in TS 23.285 [17], between nearby UEs, using E-UTRA technology but not traversing any network node.

3.2 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

AS	Access Stratum
ATG	Air To Ground
CAG	Closed Access Group
CAG-ID	Closed Access Group Identifier
CMAS	Commercial Mobile Alert System
CN	Core Network
DCI	Downlink Control Information
DRX	Discontinuous Reception
DTX	Discontinuous Transmission
eDRX	Extended DRX
ETWS	Earthquake and Tsunami Warning System
E-UTRA	Evolved UMTS Terrestrial Radio Access
E-UTRAN	Evolved UMTS Terrestrial Radio Access Network
GIN	Group ID for Network selection
H-SFN	Hyper System Frame Number
HRNN	Human-Readable Network Name
HSDN	High Speed Dedicated Network
IAB	Integrated Access and Backhaul
IMSI	International Mobile Subscriber Identity
ISA	Intended Service Area
L2	Layer-2
LO	LP-WUS Occasion
LP-WUS	Low Power-Wake Up Signal
LR	Low Power-Wake Up Receiver
MBS	Multicast/Broadcast Services
MBS FSAI	MBS Frequency Selection Area Identity
MCC	Mobile Country Code
MCCH	MBS Control Channel
MICO	Mobile Initiated Connection Only
MR	Main Receiver
MRB	MBS Radio Bearer
MTCH	MBS Traffic Channel
NAS	Non-Access Stratum
NCR	Network-Controlled Repeater
NCR-Fwd	NCR Forwarding
NCR-MT	NCR Mobile Termination
NES	Network Energy Savings
NID	Network Identifier
NPN	Non-Public Network
NR	NR Radio Access
NSAG	Network Slice AS Group
NTN	Non-Terrestrial Network
OD-SIB1	On-demand SIB1
PEI	Paging Early Indication
PEI-O	Paging Early Indication-Occasion
PH	Paging Hyperframe
PLMN	Public Land Mobile Network
PTW	Paging Time Window
RAT	Radio Access Technology
RNA	RAN-based Notification Area
RNAU	RAN-based Notification Area Update
RRC	Radio Resource Control
SDT	Small Data Transmission
SL	Sidelink
SNPN	Stand-alone Non-Public Network

TN	Terrestrial Network
TRS	Tracking Reference Signal
U2N	UE-to-Network
U2U	UE-to-UE
UAC	Unified Access Control
UE	User Equipment
UMTS	Universal Mobile Telecommunications System
VSAT	Very Small Aperture Terminal
V2X	Vehicle to Everything

4 General description of RRC_IDLE state and RRC_INACTIVE state

4.1 Overview

The RRC_IDLE state and RRC_INACTIVE state tasks can be subdivided into three processes:

- PLMN selection (for UE not operating in SNPN access mode) or SNPN selection (for UE operating in SNPN access mode);
- Cell selection and reselection;
- Location registration and RNA update.

PLMN selection, SNPN selection, cell reselection procedures, and location registration are common for both RRC_IDLE state and RRC_INACTIVE state. RNA update is only applicable for RRC_INACTIVE state. When UE selects a new PLMN or SNPN, UE transitions from RRC_INACTIVE to RRC_IDLE, as specified in TS 24.501 [14].

When a UE is switched on, a public land mobile network (PLMN) or a SNPN is selected by NAS. For the selected PLMN/SNPN, associated RAT(s) may be set, as specified in TS 23.122 [9]. The NAS shall provide a list of equivalent PLMNs or a list of equivalent SNPNs, if available, that the AS shall use for cell selection and cell reselection.

With cell selection, the UE searches for a suitable cell of the selected PLMN or selected SNPN, chooses that cell to provide available services, and monitors its control channel. This procedure is defined as "camping on the cell".

The UE shall, if necessary, then register its presence, by means of a NAS registration procedure, in the tracking area of the chosen cell. As an outcome of a successful Location Registration, the selected PLMN/SNPN then becomes the registered PLMN/SNPN, as specified in TS 23.122 [9].

If the UE finds a more suitable cell, according to the cell reselection criteria, it reselects onto that cell and camps on it. If the new cell does not belong to at least one tracking area to which the UE is registered, location registration is performed. In RRC_INACTIVE state, if the new cell does not belong to the configured RNA, an RNA update procedure is performed.

If necessary, the UE shall search for higher priority PLMNs at regular time intervals as described in TS 23.122 [9] and search for a suitable cell if another PLMN has been selected by NAS.

For UE not operating in SNPN access mode, search of available CAGs may be triggered by NAS to support manual CAG selection. The AS shall report available CAG-ID(s) together with their HRNN (if broadcast) and PLMN(s) to the NAS.

NAS may also provide the network slice(s) and Network Slice AS Group (NSAG) information, which contains NSAG(s), their applicable TA(s) if present and their priorities, to be considered by the UE during cell reselection (as specified in TS 23.501 [10], TS 24.501 [14]).

If the UE loses coverage of the registered PLMN/SNPN, either a new PLMN/SNPN is selected automatically (automatic mode), or an indication of available PLMNs/SNPNs is given to the user so that a manual selection can be performed (manual mode). As part of manual SNPN selection, the AS shall report available SNPN identifiers together with their HRNN (if broadcast) to the NAS.

Registration is not performed by UEs only capable of services that need no registration.

The UE may perform NR sidelink communication and/or V2X sidelink communication while in-coverage or out-of-coverage for sidelink, as specified in clause 8.

The U2N Remote UE, the U2N Relay UE, the U2U Remote UE, or the U2U Relay UE may perform sidelink discovery transmissions while in-coverage for the purposes of sidelink relay operations, as specified in clause 8. In addition, the U2N Remote UE, the U2U Remote UE, or the U2U Relay UE can also perform sidelink discovery transmissions while out-of-coverage for the purposes of sidelink relay operations.

An L2 U2N Remote UE in RRC_IDLE or in RRC_INACTIVE may perform all the relevant procedures (e.g., acquiring system information and paging message) via the L2 U2N Relay UE. An L2 U2N Remote UE may choose not to perform any procedures related to cell selection and reselection.

The UE may perform NR sidelink discovery transmissions while in-coverage or out-of-coverage for the purpose of sidelink non-relay operations, as specified in clause 8.

The UE may perform ranging/sidelink positioning while in-coverage or out-of-coverage, as specified in clause 8.

The purpose of camping on a cell in RRC_IDLE state and RRC_INACTIVE state is as follows:

a) It enables the UE to receive system information from the PLMN or the SNPN.

b) When registered and if the UE wishes to establish an RRC connection or resume a suspended RRC connection, it can do this by initially accessing the network on the control channel of the cell on which it is camped.

c) If the network needs to send a message or deliver data to the registered UE, it knows (in most cases) the set of tracking areas (in RRC_IDLE state) or RNA (in RRC_INACTIVE state) in which the UE is camped. It can then send a "paging" message for the UE on the control channels of all the cells in the corresponding set of areas. The UE will then receive the paging message and can respond.

d) It enables the UE to receive ETWS and CMAS notifications.

e) It enables the UE to receive MBS broadcast services.

f) It enables the UE to receive MBS multicast services in RRC_INACTIVE state.

When the UE is in RRC_IDLE state, upper layers may deactivate AS layer when MICO mode is activated as specified in TS 24.501 [14]. When MICO mode is activated, the AS configuration (e.g. priorities provided by dedicated signalling) is kept and all running timers continue to run but the UE need not perform any idle mode tasks. If a timer expires while MICO mode is activated it is up to the UE implementation whether it performs the corresponding action immediately or the latest when MICO mode is deactivated. When MICO mode is deactivated, the UE shall perform all idle mode tasks.

4.2 Functional division between AS and NAS in RRC_IDLE state and RRC_INACTIVE state

Table 4.2-1 presents the functional division between UE non-access stratum (NAS) and UE access stratum (AS) in RRC_IDLE state and RRC_INACTIVE states. The NAS part is specified in TS 23.122 [9] and the AS part in the present document.

Table 4.2-1: Functional division between AS and NAS in RRC_IDLE state and RRC_INACTIVE state

RRC_IDLE and RRC_INACTIVE state Process	UE Non-Access Stratum	UE Access Stratum
PLMN Selection and SNPN Selection	**For a UE not operating in SNPN access mode, perform the following:** Maintain a list of PLMNs in priority order according to TS 23.122 [9]. Select a PLMN using automatic or manual mode as specified in TS 23.122 [9] and request AS to select a cell belonging to this PLMN. For each PLMN, associated RAT(s) may be set. Evaluate reports of available PLMNs and any associated CAG-IDs from AS for PLMN selection. Maintain a list of equivalent PLMN identities. Maintain a list of "PLMNs not allowed to operate at the present UE location". Maintain applicable disaster roaming information for available PLMNs including potential disaster PLMNs for available PLMNs. To support manual CAG selection, provide request to search for available CAGs and evaluate reports of available CAGs from AS for CAG selection. **For a UE operating in SNPN access mode, perform the following:** Maintain a list of SNPNs according to TS 23.122 [9]. Select a SNPN using automatic or manual mode as specified in TS 23.122 [9] and request AS to select a cell belonging to this SNPN. Evaluate reports of available SNPNs from AS for SNPN selection. Maintain a list of equivalent SNPN identities.	For a UE not operating in SNPN access mode, search for available PLMNs. If associated RAT(s) is (are) set for the PLMN, search in this (these) RAT(s) and other RAT(s) for that PLMN as specified in TS 23.122 [9]. For a UE operating in SNPN access mode, search for available SNPNs only consider NR cells. Perform measurements to support PLMN/SNPN selection. Synchronise to a broadcast channel to identify found PLMNs/SNPNs. Report available PLMNs and any associated CAG-IDs with associated RAT(s) to NAS on request from NAS or autonomously. Report applicable disaster roaming information for available PLMNs autonomously including potential disaster PLMNs. For a UE operating in SNPN access mode, report available SNPNs to NAS autonomously; report information related to SNPN access with subscription of a different Credentials Holder, indicator whether onboarding is enabled, and the list of supported GINs to NAS autonomously, as specified in TS 38.331 [3]. **To support manual CAG selection, perform the following:** Search for cells broadcasting a CAG-ID. Read the HRNN (if broadcast) for each CAG-ID if a cell broadcasting a CAG-ID is found. Report CAG-ID(s) of found cell(s) broadcasting a CAG-ID together with the associated manual CAG selection allowed indicator, HRNN and PLMNto NAS. On selection of a CAG by NAS, select any acceptable or suitable cell belonging to the selected CAG and give an indication to NAS that access is possible (for the registration procedure) To support manual SNPN selection, report available SNPNs together with associated HRNNs (if available) to NAS on request from NAS.
Cell Selection	Control cell selection for example by indicating RAT(s) associated with the selected PLMN to be used initially in the search of a cell in the cell	Perform measurements needed to support cell selection.

RRC_IDLE and RRC_INACTIVE state Process	UE Non-Access Stratum	UE Access Stratum
	selection. For a UE not operating in SNPN access mode, maintain a list of equivalent PLMN identities and provide the list to AS. Maintain a list of "Forbidden Tracking Areas" and provide the list to AS. Maintain a list of "PLMNs not allowed to operate at the present UE location" and provide the list to AS. Maintain a list of "PLMNs with associated access technology restrictions" and provide the list to AS. For a UE not operating in SNPN access mode: Maintain Allowed CAG list and optional CAG-only indication along with associated PLMN ID(s) on which the UE is allowed access and provide these lists to AS. To support manual CAG selection, select a CAG and request AS to select a cell belonging to this CAG.	Detect and synchronise to a broadcast channel. Receive and handle broadcast information. Forward NAS system information to NAS. Search for a suitable cell. The cells broadcast one or more 'PLMN identity' or 'SNPN identity' (for a UE operating in SNPN access mode) in the system information. Respond to NAS whether such cell is found or not. If associated RATs is (are) set for the PLMN, perform the search in this (these) RAT(s) and other RATs for that PLMN as specified in TS 23.122 [9]. If a cell is found which satisfies cell selection criteria, camp on that cell.

RRC_IDLE and RRC_INACTIVE state Process	UE Non-Access Stratum	UE Access Stratum
Cell Reselection	For a UE not operating in SNPN access mode, maintain a list of equivalent PLMN identities and provide the list to AS. For a UE operating in SNPN access mode, maintain a list of equivalent SNPN identities and provide the list to AS. Maintain a list of "Forbidden Tracking Areas" and provide the list to AS. Maintain a list of "PLMNs not allowed to operate at the present UE location" and provide the list to AS. Maintain a list of "PLMNs with associated access technology restrictions" and provide the list to AS. For a UE not operating in SNPN access mode, maintain Allowed CAG list and optional CAG-only indication along with associated PLMN ID(s) on which the UE is allowed access and provide these lists to AS. Maintain the network slice(s) and NSAG information and provide network slice(s) and NSAG information to AS.	Perform measurements needed to support cell reselection. Detect and synchronise to a broadcast channel. Receive and handle broadcast information. Forward NAS system information to NAS. Change cell if a more suitable cell is found. Derive cell reselection priorities for slice-based cell reselection.
Location registration	Register the UE as active after power on. Register the UE's presence in a registration area, for instance regularly or when entering a new tracking area. Deregister UE when shutting down. Maintain a list of "Forbidden Tracking Areas". Maintain a list of "PLMNs not allowed to operate at the present UE location". Control and restrict location registration for a UE in eCall Only Mode.	Report registration area information to NAS.
RAN Notification Area Update	Not applicable.	Register the UE's presence in a RAN-based notification area (RNA), periodically or when entering a new RNA.

4.3 Service types in RRC_IDLE state

This clause defines the level of service that may be provided by the network to a UE in RRC_IDLE state. The following three levels of services are provided while a UE is in RRC_IDLE state:

- Limited service (emergency calls, ETWS and CMAS on an acceptable cell);
- Normal service (for public use or non-public use on a suitable cell);
- Operator service (for operators only on a reserved cell).

4.4 Service types in RRC_INACTIVE state

This clause defines the level of service that may be provided by the network to a UE in RRC_INACTIVE state. The following two levels of services are provided while a UE is in RRC_INACTIVE state:

- Normal service (for public use or non-public use on a suitable cell);
- Operator service (for operators only on a reserved cell).

4.5 Cell Categories

The cells are categorised according to which services they offer:

acceptable cell:

An "acceptable cell" is a cell on which the UE may camp to obtain limited service (originate emergency calls and receive ETWS and CMAS notifications). Such a cell shall fulfil the following requirements, which is the minimum set of requirements to initiate an emergency call and to receive ETWS and CMAS notification in an NR network:

- The cell is not barred, see clause 5.3.1;
- The cell selection criteria are fulfilled, see clause 5.2.3.2.

suitable cell:

For UE not operating in SNPN Access Mode, a cell is considered as suitable if the following conditions are fulfilled:

- The cell is part of either the selected PLMN or the registered PLMN or PLMN of the Equivalent PLMN list, and for that PLMN either:
  - The PLMN-ID of that PLMN is broadcast by the cell with no associated CAG-IDs and CAG-only indication in the UE for that PLMN (TS 23.501 [10]) is absent or false;
  - Allowed CAG list in the UE for that PLMN (TS 23.501 [10]) includes a CAG-ID broadcast by the cell for that PLMN;
- The cell selection criteria are fulfilled, see clause 5.2.3.2.

According to the latest information provided by NAS:

- The cell is not barred, see clause 5.3.1;
- The cell is part of at least one TA that is not part of the list of "Forbidden Tracking Areas for Roaming" (TS 22.011 [18]), which belongs to a PLMN that fulfils the first bullet above;
- The access technology of the cell is not marked as restricted for the PLMN that fulfils the first bullet above in the list of "PLMNs with associated access technology restrictions" (TS 23.122 [9]).

For UE operating in SNPN Access Mode, a cell is considered as suitable if the following conditions are fulfilled:

- The cell is part of the selected SNPN or the registered SNPN or SNPN of the Equivalent SNPN list of the UE;
- The cell selection criteria are fulfilled, see clause 5.2.3.2;

According to the latest information provided by NAS:

- The cell is not barred, see clause 5.3.1;
- The cell is part of at least one TA that is not part of the list of "Forbidden Tracking Areas for Roaming" which belongs to the selected SNPN or the registered SNPN or SNPN of the Equivalent SNPN list of the UE.

barred cell:

A cell is barred if it is so indicated in the system information, as specified in TS 38.331 [3].

reserved cell:

A cell is reserved if it is so indicated in system information, as specified in TS 38.331 [3].

Following exception to these definitions are applicable for UEs:

- if a UE has an ongoing emergency call, all acceptable cells of that PLMN/SNPN are treated as suitable for the duration of the emergency call.
- camped on a cell that belongs to a tracking area that is forbidden for regional provision of service; a cell that belongs to a tracking area that is forbidden for regional provision service (TS 23.122 [9], TS 24.501 [14]) is suitable but provides only limited service.
- if the UE in RRC_IDLE fulfils the conditions to support NR sidelink communication/discovery or V2X sidelink communication in limited service state as specified in TS23.287 [16] clause 5.7, the UE may perform NR sidelink communication/discovery or V2X sidelink communication.

NOTE: UE is not required to support manual search and selection of PLMN or CAG or SNPN while in RRC CONNECTED state. The UE may use local release of RRC connection to perform manual search if it is not possible to perform the search while RRC connected.

5 Process and procedure descriptions

5.1 PLMN selection and SNPN selection

In the UE not operating in SNPN access mode, the AS shall report available PLMNs and any associated CAG-IDs to the NAS on request from the NAS or autonomously. In the UE operating in SNPN access mode, the AS shall report available SNPNs to the NAS on request from the NAS or autonomously.

During PLMN selection, based on the list of PLMN identities in priority order, the particular PLMN may be selected either automatically or manually. Each PLMN in the list of PLMN identities is identified by a 'PLMN identity'. In the system information on the broadcast channel, the UE can receive one or multiple 'PLMN identity' in a given cell. The result of the PLMN selection performed by NAS (see TS 23.122 [9]) is an identifier of the selected PLMN.

During SNPN selection, based on the list of SNPN identities, the particular SNPN may be selected either automatically or manually. Each SNPN in the list of SNPN identities is identified by a 'SNPN identity'. In the system information on the broadcast channel, the UE can receive one or multiple 'SNPN identity' in a given cell and optionally may receive associated HRNNs; the UE may also optionally receive indicators for whether an SNPN allows access using credentials from a Credentials Holder, whether an SNPN allows registration attempts from UEs that are not explicitly configured to select this SNPN, and whether an SNPN allows onboarding; the UE may also optionally receive a list of supported Group IDs for Network selection (see TS 38.331 [3]). The result of the SNPN selection performed by NAS (see TS 23.122 [9]) is an identifier of the selected SNPN.

5.1.1 Support for PLMN selection

5.1.1.1 General

On request of the NAS, the AS shall perform a search for available PLMNs and report them to NAS.

5.1.1.2 NR case

The UE shall scan all RF channels in the NR bands according to its capabilities to find available PLMNs and available CAGs. On each carrier, the UE shall search for the strongest cell and read its system information, in order to find out which PLMN(s) the cell belongs to and any associated CAG(s). For operation with shared spectrum channel access, the UE may also read the system information of multiple strongest cell(s). If the UE can read one or several PLMN identities in the strongest cell or the multiple strongest cell(s) in case of operation with shared spectrum channel access, each found PLMN (see the PLMN reading in TS 38.331 [3]) shall be reported to the NAS as a high quality PLMN (but without the RSRP value) and any associated CAG-ID, provided that the following high-quality criterion is fulfilled:

1. For an NR cell, the measured RSRP value shall be greater than or equal to -110 dBm.

Found PLMNs that do not satisfy the high-quality criterion but for which the UE has been able to read the PLMN identities are reported to the NAS together with their corresponding RSRP values and any associated CAG-ID. The quality measure reported by the UE to NAS shall be the same for each PLMN found in one cell.

The search for PLMNs may be stopped on request from the NAS. The UE may optimise PLMN search by using stored information e.g. frequencies and optionally also information on cell parameters from previously received measurement control information elements.

Once the UE has selected a PLMN, the cell selection procedure shall be performed in order to select a suitable cell of that PLMN to camp on.

To support manual CAG selection, the UE shall upon request by NAS report available CAG-ID(s) together with their manual CAG selection allowed indicator (if broadcast), HRNN (if broadcast) and PLMN(s) to the NAS. If NAS has selected a CAG and provided this selection to AS, the UE shall search for an acceptable or suitable cell belonging to the selected CAG to camp on.

5.1.1.3 E-UTRA case

Support for PLMN selection in E-UTRA is described in TS 36.304 [7].

5.1.2 Support for SNPN selection

5.1.2.1 General

On request of the NAS, the AS shall perform a search for available SNPNs on only NR cells and report them to NAS.

5.1.2.2 NR case

The UE shall scan all RF channels in the NR bands according to its capabilities to find available SNPNs. On each carrier, the UE shall search for the strongest cell and read its system information, in order to find out which SNPN(s) the cell belongs to. For operation with shared spectrum channel access, the UE may also read the system information of multiple strongest cell(s). If the UE can read one or several SNPN identities in the strongest cell, each found SNPN (see the SNPN reading in TS 38.331 [3]) shall be reported to the NAS. For manual selection, UE shall upon request by NAS report available SNPN identifiers together with their HRNN (if broadcast) to the NAS and the search for available SNPNs may be stopped on request of the NAS.

The search for SNPNs may be stopped on request from the NAS. The UE may optimise SNPN search by using stored information e.g. frequencies and optionally also information on cell parameters from previously received measurement control information elements.

Once the UE has selected a SNPN, the cell selection procedure shall be performed in order to select a suitable cell of that SNPN to camp on.

5.2 Cell selection and reselection

5.2.1 Introduction

UE shall perform measurements for cell selection and reselection purposes as specified in TS 38.133 [8].

When evaluating Srxlev and Squal of non-serving cells for reselection evaluation purposes, the UE shall use parameters provided by the serving cell and for the final check on cell selection criterion, the UE shall use parameters provided by the target cell for cell reselection.

The NAS can control the RAT(s) in which the cell selection should be performed, for instance by indicating RAT(s) associated with the selected PLMN, and by maintaining a list of forbidden registration area(s), and by maintaining a list of "PLMNs with associated access technology restrictions" and a list of equivalent PLMNs. The UE shall select a suitable cell based on RRC_IDLE or RRC_INACTIVE state measurements and cell selection criteria.

In order to expedite the cell selection process, stored information for several RATs, if available, may be used by the UE.

When camped on a cell, the UE shall regularly search for a better cell according to the cell reselection criteria. If a better cell is found, that cell is selected. The change of cell may imply a change of RAT. Details on performance requirements for cell reselection can be found in TS 38.133 [8].

For NCRs, if the NCR-MT in RRC_INACTIVE (re)selects a cell other than the last serving cell on which side control configuration was received, then the NCR-MT shall indicate to NCR-Fwd to cease forwarding. If the NCR-MT in RRC_INACTIVE detects no suitable cell, then the NCR-MT shall indicate to NCR-Fwd to cease forwarding.

The NAS is informed if the cell selection and reselection result in changes in the received system information relevant for NAS.

For normal service, the UE shall camp on a suitable cell, monitor control channel(s) of that cell so that the UE can:

- receive system information from the PLMN or SNPN; and
  - receive registration area information from the PLMN or SNPN, e.g., tracking area information; and
  - receive other AS and NAS Information; and
- if registered:
  - receive paging and notification messages from the PLMN or SNPN; and
  - initiate transfer to Connected mode.

For cell selection in multi-beam operations, measurement quantity of a cell is up to UE implementation.

For cell reselection in multi-beam operations, including inter-RAT reselection from E-UTRA to NR, the measurement quantity of this cell is derived amongst the beams corresponding to the same cell based on SS/PBCH block as follows:

- if *nrofSS-BlocksToAverage* (*maxRS-IndexCellQual* in E-UTRA) is not configured in *SIB2/SIB4* (*SIB24* in E-UTRA); or
- if *absThreshSS-BlocksConsolidation* (*threshRS-Index* in E-UTRA) is not configured in *SIB2/SIB4* (*SIB24* in E-UTRA); or
- if the highest beam measurement quantity value is below or equal to *absThreshSS-BlocksConsolidation* (*threshRS-Index* in E-UTRA):
  - derive a cell measurement quantity as the highest beam measurement quantity value, where each beam measurement quantity is described in TS 38.215 [11].
- else:
  - derive a cell measurement quantity as the linear average of the power values of up to *nrofSS-BlocksToAverage* (*maxRS-IndexCellQual* in E-UTRA) of highest beam measurement quantity values above *absThreshSS-BlocksConsolidation* (*threshRS-Index* in E-UTRA).

NOTE: If both suitable cell(s) and suitable L2 U2N Relay UE(s) (as specified in TS 38.331 [3]) are available, it is up to L2 U2N Remote UE's implementation to select either a suitable cell or a suitable L2 U2N Relay UE.

5.2.2 States and state transitions in RRC_IDLE state and RRC_INACTIVE state

Figure 5.2.2-1 shows the states and state transitions and procedures in RRC_IDLE and RRC_INACTIVE. Whenever a new PLMN selection or new SNPN selection is performed, it causes an exit to number 1.

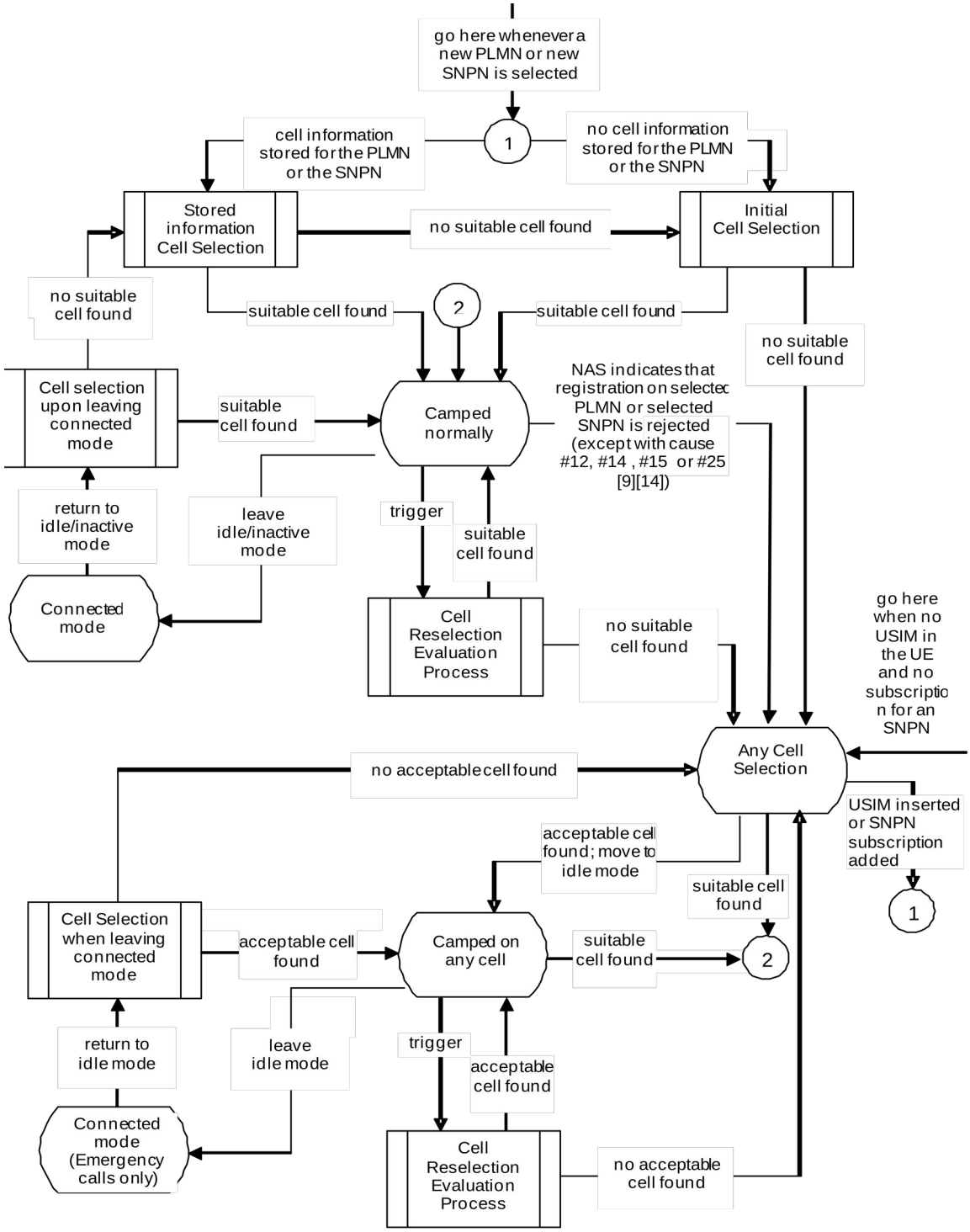


Figure 5.2.2-1: RRC_IDLE and RRC_INACTIVE Cell Selection and Reselection

5.2.3 Cell Selection process

5.2.3.1 Description

Cell selection is performed by one of the following two procedures:

a) Initial cell selection (no prior knowledge of which RF channels are NR frequencies):

1. The UE shall scan all RF channels in the NR bands according to its capabilities to find a suitable cell.
2. On each frequency, the UE need only search for the strongest cell, except for operation with shared spectrum channel access where the UE may search for the next strongest cell(s).
3. Once a suitable cell is found, this cell shall be selected.

b) Cell selection by leveraging stored information:

1. This procedure requires stored information of frequencies and optionally also information on cell parameters from previously received measurement control information elements or from previously detected cells.
2. Once the UE has found a suitable cell, the UE shall select it.
3. If no suitable cell is found, the initial cell selection procedure in a) shall be started.

NOTE: Priorities between different frequencies or RATs provided to the UE by system information or dedicated signalling are not used in the cell selection process.

5.2.3.2 Cell Selection Criterion

The cell selection criterion S is fulfilled when:

$Srxlev > 0$ AND $Squal > 0$

where:

$Srxlev = Q_{rxlevmeas} - (Q_{rxlevmin} + Q_{rxlevminoffset}) - P_{compensation} - Qoffset_{temp}$

$Squal = Q_{qualmeas} - (Q_{qualmin} + Q_{qualminoffset}) - Qoffset_{temp}$

where:

Srxlev	Cell selection RX level value (dB)
Squal	Cell selection quality value (dB)
$Qoffset_{temp}$	Offset temporarily applied to a cell as specified in TS 38.331 [3] (dB)
$Q_{rxlevmeas}$	Measured cell RX level value (RSRP)
$Q_{qualmeas}$	Measured cell quality value (RSRQ)
$Q_{rxlevmin}$	Minimum required RX level in the cell (dBm). If the UE supports SUL frequency for this cell, $Q_{rxlevmin}$ is obtained from *q-RxLevMinSUL*, if present, in *SIB1*, *SIB2* and *SIB4*, additionally, if $Q_{rxlevminoffsetcellSUL}$ is present in *SIB3* and *SIB4* for the concerned cell, this cell specific offset is added to the corresponding Qrxlevmin to achieve the required minimum RX level in the concerned cell; else $Q_{rxlevmin}$ is obtained from *q-RxLevMin* in *SIB1, SIB2* and *SIB4*, additionally, if $Q_{rxlevminoffsetcell}$ is present in *SIB3* and *SIB4* for the concerned cell, this cell specific offset is added to the corresponding Qrxlevmin to achieve the required minimum RX level in the concerned cell.
$Q_{qualmin}$	Minimum required quality level in the cell (dB). Additionally, if $Q_{qualminoffsetcell}$ is signalled for the concerned cell, this cell specific offset is added to achieve the required minimum quality level in the concerned cell.
$Q_{rxlevminoffset}$	Offset to the signalled $Q_{rxlevmin}$ taken into account in the Srxlev evaluation as a result of a periodic search for a higher priority PLMN while camped normally in a VPLMN, as specified in TS 23.122 [9].
$Q_{qualminoffset}$	Offset to the signalled $Q_{qualmin}$ taken into account in the Squal evaluation as a result of a periodic search for a higher priority PLMN while camped normally in a VPLMN, as specified in TS 23.122 [9].
$P_{compensation}$	For FR1, if the UE supports the *additionalPmax* in the *NR-NS-PmaxList*, if present, in *SIB1, SIB2* and *SIB4:* $max(P_{EMAX1} - P_{PowerClass}, 0) - (min(P_{EMAX2}, P_{PowerClass}) - min(P_{EMAX1}, P_{PowerClass}))$ *(dB);* *else:* $max(P_{EMAX1} - P_{PowerClass}, 0)$ *(dB)* For FR2, $P_{compensation}$ is set to 0. For IAB-MT, $P_{compensation}$ is set to 0.
P_{EMAX1}, P_{EMAX2}	Maximum TX power level of a UE may use when transmitting on the uplink in the cell (dBm) defined as P_{EMAX} in TS 38.101 [15]. If UE supports SUL frequency for this cell, P_{EMAX1} and P_{EMAX2} are obtained from the *p-Max* for SUL in *SIB1* and *NR-NS-PmaxList* for SUL respectively in *SIB1, SIB2* and *SIB4* as specified in TS 38.331 [3], else P_{EMAX1} and P_{EMAX2} are obtained from the *p-Max* and *NR-NS-PmaxList* respectively in *SIB1*, *SIB2* and *SIB4* for normal UL as specified in TS 38.331 [3].
$P_{PowerClass}$	Maximum RF output power of the UE (dBm) according to the UE power class as defined in TS 38.101-1 [15].

The signalled values $Q_{rxlevminoffset}$ and $Q_{qualminoffset}$ are only applied when a cell is evaluated for cell selection as a result of a periodic search for a higher priority PLMN while camped normally in a VPLMN (TS 23.122 [9]). During this periodic search for higher priority PLMN, the UE may check the S criteria of a cell using parameter values stored from a different cell of this higher priority PLMN.

5.2.3.3 E-UTRAN case in Cell Selection

The cell selection criteria and procedures in E-UTRAN are specified in TS 36.304 [7].

5.2.4 Cell Reselection evaluation process

5.2.4.1 Reselection priorities handling

Absolute priorities of different NR frequencies or inter-RAT frequencies may be provided to the UE in the system information, in the *RRCRelease* message, or by inheriting from another RAT at inter-RAT cell (re)selection. In the case of system information, an NR frequency or inter-RAT frequency may be listed without providing a priority (i.e. the field *cellReselectionPriority* is absent for that frequency). If any fields with *cellReselectionPriority* or *nsag-*

CellReselectionPriority are provided in dedicated signalling, the UE shall ignore any fields with *cellReselectionPriority* and *nsag-CellReselectionPriority* provided in system information.

When UE is in camped normally state, if it supports slice-based cell reselection and has received the network slice(s) and NSAG information from NAS to be used for cell reselection, UE shall derive reselection priorities according to clause 5.2.4.11.

NOTE 00: UE derives reselection priorities according to clause 5.2.4.11 also in case *SIB16* (see TS 38.331 [3]) is not broadcast in the camped cell.

If dedicated frequency priority parameters (*od-sib1-CellReselectionPriority*, *od-sib1-CellReselectionSubPriority*) are provided in system information, the UE supporting OD-SIB1 ignores the *cellReselectionPriority* and *cellReselectionSubPriority* in the system information and applies the dedicated ones to determine frequency prioritization. If *intraFreqOD-SIB1-ExcludedCellList* is provided in system information, the UE supporting OD-SIB1 ignores *intraFreqExcludedCellList* and does not consider the cell(s) (if any) in *intraFreqOD-SIB1-ExcludedCellList* as candidates for cell reselection. If *interFreqOD-SIB1-ExcludedCellList* is provided in system information, the UE supporting OD-SIB1 ignores *interFreqExcludedCellList* and does not consider the cell(s) (if any) in *interFreqOD-SIB1-ExcludedCellList* as candidates for cell reselection.

If UE is in *camped on any cell* state, UE shall only apply the priorities provided by system information from current cell, and the UE preserves priorities provided by dedicated signalling and *deprioritisationReq* received in *RRCRelease* unless specified otherwise. When the UE in camped normally state, has only dedicated priorities other than for the current frequency, the UE shall consider the current frequency to be the lowest priority frequency (i.e. lower than any of the network configured values). When the HSDN capable UE is in High-mobility state, the UE shall always consider the HSDN cells to be the highest priority (i.e., higher than any other network configured priorities). When the HSDN capable UE is not in High-mobility state, the UE shall always consider HSDN cells to be the lowest priority (i.e., lower than any other network configured priorities). The Aerial UE may consider the frequency to be the highest priority (i.e., higher than any other network configured priorities) if the frequency is configured with *uav-PrioritizedFrequency* without *uav-PrioritizedFrequencyAltitudeRange*. The Aerial UE may consider the frequency to be the highest priority if the frequency is configured with *uav-PrioritizedFrequency* and *uav-PrioritizedFrequencyAltitudeRange*, and if the Aerial UE is within the corresponding altitude range. If the UE is configured to perform both NR sidelink communication and V2X sidelink communication, the UE may consider the frequency providing both NR sidelink communication configuration and V2X sidelink communication configuration to be the highest priority. If the UE is configured to perform NR sidelink communication and not perform V2X communication, the UE may consider the frequency providing NR sidelink communication configuration to be the highest priority. If the UE is configured to perform V2X sidelink communication and not perform NR sidelink communication, the UE may consider the frequency providing V2X sidelink communication configuration to be the highest priority. If the UE is configured to perform ranging/sidelink positioning, the UE may consider the frequency providing ranging/sidelink positioning configuration to be the highest priority.

A UE on a vehicle with a mobile-IAB cell may consider the frequency for which a mobile-IAB cell is the best cell to be the highest priority. The UE identifies a mobile-IAB cell by *mobileIAB-Cell* in SIB1 (see TS 38.331 [3]). The UE may narrow its search scope for mobile-IAB cell(s) by *mobileIAB-CellList* if broadcasted in SIB4 (see TS 38.331 [3]). A non-mobile-IAB cell may be excluded from this mobile IAB frequency prioritization for up to 300 seconds.

NOTE 0a: The frequency only providing the anchor frequency configuration should not be prioritized for V2X service during cell reselection, as specified in TS 38.331[3].

NOTE 0b: When UE is configured to perform NR sidelink communication or V2X sidelink communication performs cell reselection, it may consider the frequencies providing the intra-carrier and inter-carrier configuration have equal priority in cell reselection.

NOTE 0c: The prioritization among the frequencies which UE considers to be the highest priority frequency is left to UE implementation unless otherwise stated.

NOTE 0d: The UE is configured to perform V2X sidelink communication or NR sidelink communication, if it has the capability and is authorized for the corresponding sidelink operation.

NOTE 0e: When UE is configured to perform both NR sidelink communication and V2X sidelink communication, but cannot find a frequency which can provide both NR sidelink communication configuration and V2X sidelink communication configuration, UE may consider the frequency providing either NR sidelink communication configuration or V2X sidelink communication configuration to be the highest priority.

NOTE 0f: Void.

NOTE 0g: How the UE determines itself to be on a vehicle with a mobile-IAB cell is left to the UE's implementation.

The UE shall only perform cell reselection evaluation for NR frequencies and inter-RAT frequencies that are given in system information and for which the UE has a priority provided.

If the MBS broadcast capable UE is receiving or interested to receive an MBS broadcast service(s) and can only receive this MBS broadcast service(s) by camping on a frequency on which it is provided; or if the MBS ISA capable UE is camping on an NTN cell where the ISA(s) of MBS broadcast services are provided, and the UE is in the ISA(s) associated with MBS broadcast service(s) it is receiving or interested to receive and can only receive this MBS broadcast service(s) by camping on a frequency on which it is provided, the UE may consider that frequency to be the highest priority during the MBS broadcast session as specified in TS 38.300 [2] as long as the two following conditions are fulfilled:

1) SIB1 scheduling information of the cell reselected by the UE due to frequency prioritization for MBS contains SIB20;

2) Either:

   - One or more MBS FSAI(s) of that frequency is indicated in SIB21 of the serving cell and the same MBS FSAI(s) is also indicated for this MBS broadcast service in MBS User Service Description (USD) as specified in TS 26.517 [20], or

   - SIB21 is not provided in the serving cell and that frequency is included in the USD of this service, or

   - SIB21 is provided in the serving cell but does not provide the frequency mapping for the concerned service, and that frequency is included in the USD of this service.

NOTE 0g: It is up to UE implementation which frequency to select, when the USD provides multiple frequencies for the service the UE is interested in.

If the MBS broadcast capable UE is receiving or interested to receive an MBS broadcast service or if the MBS ISA capable UE is camping on an NTN cell where the ISA(s) of MBS broadcast services area are provided, and the UE is outside of the ISA(s) associated with MBS broadcast service(s) it is receiving or interested to receive, the UE may consider cell reselection candidate frequencies at which it cannot receive the MBS broadcast service to be of the lowest priority during the MBS broadcast session as specified in TS 38.300 [2], as long as SIB1 scheduling information of the cell contains SIB20 on the MBS frequency which the UE monitors and as long as the condition 2) above is fulfilled for the serving cell.

NOTE 0h: Example scenarios in which such down-prioritisation may be needed include the cases where camping is not possible for the UE on the MBS broadcast frequency (e.g. the MBS broadcast frequency belongs to a PLMN different from UE's registered PLMN) while the UE can receive the MBS broadcast service when camped on another frequency than the MBS broadcast frequency or current frequency.

NOTE 0i: The frequency prioritization for MBS broadcast, NR sidelink communication, or V2X sidelink communication may override the re-selection priorities for slice-based cell reselection.

NOTE 0j: The ISA(s) can be ISA(s) provided in *SIB27* and/or target service area in USD. It is up to UE's implementation to decide whether it is inside ISA(s) or not.

In case UE receives *RRCRelease* with *deprioritisationReq*, UE shall consider current frequency and stored frequencies due to the previously received *RRCRelease* with *deprioritisationReq* or all the frequencies of NR to be the lowest priority frequency (i.e. lower than any of the network configured values) while T325 is running irrespective of camped RAT. The UE shall delete the stored deprioritisation request(s) when a PLMN selection or SNPN selection is performed on request by NAS (TS 23.122 [9]).

NOTE 1: UE should search for a higher priority layer for cell reselection as soon as possible after the change of priority. The minimum related performance requirements specified in TS 38.133 [8] are still applicable.

NOTE 1a: The UE does not consider MBS broadcast, NR sidelink communication or V2X sidelink communication functionality to replace cell reselection priorities caused by HSDN, UAV or *deprioritisationReq* functionality.

The UE shall delete priorities provided by dedicated signalling when:

- the UE enters a different RRC state; or
- the optional validity time of dedicated priorities (T320) expires; or
- the UE receives an *RRCRelease* message with the field *cellReselectionPriorities* absent; or
- a PLMN selection or SNPN selection is performed on request by NAS (TS 23.122 [9]).

NOTE 2: Equal priorities between RATs are not supported.

The UE shall not consider any exclude-listed cells as candidate for cell reselection.

The UE shall consider only the allow-listed cells, if configured, as candidates for cell reselection.

NCR-MT may be configured with additional allowed cell list and/or forbidden cell list, see TS 38.300 [2]. The NCR-MT shall consider only the allowed cell list, if configured by OAM as candidates for cell reselection (ignore above exclude-listed cells and/or allow-listed cells). The NCR-MT shall not consider the cells for cell reselection in the forbidden cell list, if configured by OAM.

The UE in RRC_IDLE state shall inherit the priorities provided by dedicated signalling and the remaining validity time (i.e. T320 in NR and E-UTRA), if configured, at inter-RAT cell (re)selection.

NOTE 3: The network may assign dedicated cell reselection priorities for frequencies not configured by system information.

5.2.4.2 Measurement rules for cell re-selection

Following rules are used by the UE to limit needed measurements:

- If the serving cell fulfils Srxlev > $S_{IntraSearchP}$ and Squal > $S_{IntraSearchQ}$:
  - If *distanceThresh* and *referenceLocation* are broadcasted in *SIB19*, and if UE supports location-based measurement initiation for NTN (quasi-)Earth-fixed cell and has obtained its location information:
    - If the distance between UE and the serving cell reference location *referenceLocation* is shorter than *distanceThresh*, the UE may not perform intra-frequency measurements;
    - Else, the UE shall perform intra-frequency measurements;
  - else if *distanceThresh* and *movingReferenceLocation* are broadcasted in *SIB19*, and if UE supports location-based measurement initiation for NTN Earth-moving cell and has obtained its location information:
    - if the distance between UE's location and the serving cell reference location determined based on *movingReferenceLocation* is shorter than *distanceThresh*, the UE may not perform intra-frequency measurements;
    - else, the UE shall perform intra-frequency measurements;
  - Else, the UE may not perform intra-frequency measurements;
- Else, the UE shall perform intra-frequency measurements.
- The UE shall apply the following rules for NR inter-frequencies and inter-RAT frequencies which are indicated in system information and for which the UE has priority provided as defined in 5.2.4.1:
  - For a NR inter-frequency or inter-RAT frequency with a reselection priority higher than the reselection priority of the current NR frequency, the UE shall perform measurements of higher priority NR inter-frequency or inter-RAT frequencies according to TS 38.133 [8].
  - For a NR inter-frequency with an equal or lower reselection priority than the reselection priority of the current NR frequency and for inter-RAT frequency with lower reselection priority than the reselection priority of the current NR frequency:
    - If the serving cell fulfils Srxlev > $S_{nonIntraSearchP}$ and Squal > $S_{nonIntraSearchQ}$:

- If *distanceThresh* and *referenceLocation* are broadcasted in *SIB19*, and if UE supports location-based measurement initiation for NTN (quasi-)Earth-fixed cell and has obtained its UE location information:
  - If the distance between UE and the serving cell reference location *referenceLocation* is shorter than *distanceThresh*, the UE may choose not to perform measurements of NR inter-frequency cells of equal or lower priority, or inter-RAT frequency cells of lower priority;
  - Else, the UE shall perform measurements of NR inter-frequency cells of equal or lower priority, or inter-RAT frequency cells of lower priority according to TS 38.133 [8];
- else if *distanceThresh* and *movingReferenceLocation* are broadcasted in *SIB19*, and if UE supports location-based measurement initiation for NTN Earth-moving cell and has obtained its location information:
  - if the distance between UE's location and the serving cell reference location determined based on *movingReferenceLocation* is shorter than *distanceThresh*, the UE may not perform measurements of NR inter-frequency cells of equal or lower priority, or inter-RAT frequency cells of lower priority;
  - else, the UE shall perform measurements of NR inter-frequency cells of equal or lower priority, or inter-RAT frequency cells of lower priority according to TS 38.133 [8];
- Else, the UE may choose not to perform measurements of NR inter-frequency cells of equal or lower priority, or inter-RAT frequency cells of lower priority;
- Else, the UE shall perform measurements of NR inter-frequency cells of equal or lower priority, or inter-RAT frequency cells of lower priority according to TS 38.133 [8].

- If the UE supports relaxed measurement and *relaxedMeasurement* is present in *SIB2*, the UE may further relax the needed measurements, as specified in clause 5.2.4.9.

- UE supporting LP-WUS in RRC_IDLE and RRC_INACTIVE states and *relaxedMeasurementForServingAndNeighboringCell* is present in *SIB2*, the UE may perform relaxed serving cell and further neighbouring cell measurements on MR as specified in clause 5.2.4.12.1. If the UE supports LP-WUS in RRC_IDLE and RRC_INACTIVE states and *offloadMeasurementForServingCell* is present in *SIB2*, the UE may perform serving cell measurement offloading from MR to LR as specified in clause 5.2.4.12.3.

- For UE camping on NTN cell, if the UE supports skipping TN measurement, and the UE has obtained its location information, and if *coverageAreaInfoList* and *tn-AreaIdList* are broadcast in system information, the UE may not perform measurements of a TN frequency when UE is not in the coverage of that frequency provided via *tn-AreaIdList*, regardless of the frequency priority.

- For Aerial UE, if *ssb-ToMeasureAltitudeBasedList-19* is configured for the measurement frequency, when the UE is within an altitude range indicated by corresponding *altitudeRange-r18* in *ssb-ToMeasureAltitudeBasedList-r19*, it may prioritize the measurements on the SSB(s) configured in *ssb-ToMeasure-r18 in ssb-ToMeasureAltitudeBasedList-r19*. Otherwise (i.e., the UE is within an altitude range indicated by corresponding *altitudeRange-r18* and *ssb-ToMeasure-r18* is absent in *ssb-ToMeasureAltitudeBasedList-r19*) it measures on all SS-blocks. When the UE is outside all the altitude ranges indicated by corresponding *altitudeRange-r18* (if any) in *ssb-ToMeasureAltitudeBasedList-r19*, *ssb-ToMeasure* (without suffix) applies.

If the *t-Service* of the serving cell is present in *SIB19*, and if UE supports time-based measurement initiation, the UE shall perform intra-frequency, inter-frequency or inter-RAT measurements before the *t-Service*, regardless of the distance between UE and the serving cell reference location or whether the serving cell fulfils Srxlev > $S_{IntraSearchP}$ and Squal > $S_{IntraSearchQ}$, or Srxlev > $S_{nonIntraSearchP}$ and Squal > $S_{nonIntraSearchQ}$, The exact time to start measurement before *t-Service* is up to UE implementation. UE shall perform measurements of higher priority NR inter-frequency or inter-RAT frequencies according to TS 38.133 [8] regardless of the remaining service time of the serving cell (i.e. time remaining until *t-Service*).

NOTE 1: When evaluating the distance between UE and the serving cell reference location, it is up to UE implementation to obtain UE location information.

NOTE 2: In the Earth-moving cell, it is up to UE implementation to maintain a valid serving cell reference location, which is derived based on the serving satellite ephemeris, *epochTime* and *movingReferenceLocation*.

5.2.4.3 Mobility states of a UE

5.2.4.3.0 Introduction

The UE mobility state is determined if the parameters (T_{CRmax}, N_{CR_H}, N_{CR_M}, $T_{CRmaxHyst}$ and *cellEquivalentSize*) are broadcasted in system information for the serving cell.

State detection criteria:

Normal-mobility state criteria:

- If number of cell reselections during time period T_{CRmax} is less than N_{CR_M}.

Medium-mobility state criteria:

- If number of cell reselections during time period T_{CRmax} is greater than or equal to N_{CR_M} but less than or equal to N_{CR_H}.

High-mobility state criteria:

- If number of cell reselections during time period T_{CRmax} is greater than N_{CR_H}.

The UE shall not consider consecutive reselections where a cell is reselected again right after one reselection for mobility state detection criteria. If the UE is capable of HSDN and the *cellEquivalentSize* is configured, the UE counts the number of cell reselections for this cell as *cellEquivalentSize* configured for this cell.

State transitions:

The UE shall:

- if the criteria for High-mobility state is detected:
  - enter High-mobility state.
- else if the criteria for Medium-mobility state is detected:
  - enter Medium-mobility state.
- else if criteria for either Medium- or High-mobility state is not detected during time period $T_{CRmaxHyst}$:
  - enter Normal-mobility state.

If the UE is in High- or Medium-mobility state, the UE shall apply the speed dependent scaling rules as defined in clause 5.2.4.3.1.

5.2.4.3.1 Scaling rules

UE shall apply the following scaling rules:

- If neither Medium- nor High-mobility state is detected:
  - no scaling is applied.
- If High-mobility state is detected:
  - Add the *sf-High* of "Speed dependent ScalingFactor for Q_{hyst}" to Q_{hyst} if broadcasted in system information;
  - For NR cells, multiply $Treselection_{NR}$ by the *sf-High* of "Speed dependent ScalingFactor for $Treselection_{NR}$" if broadcasted in system information;
  - For EUTRA cells, multiply $Treselection_{EUTRA}$ by the *sf-High* of "Speed dependent ScalingFactor for $Treselection_{EUTRA}$" if broadcasted in system information.

- If Medium-mobility state is detected:
  - Add the *sf-Medium* of "Speed dependent ScalingFactor for Q_{hyst}" to Q_{hyst} if broadcasted in system information;
  - For NR cells, multiply $Treselection_{NR}$ by the *sf-Medium* of "Speed dependent ScalingFactor for $Treselection_{NR}$" if broadcasted in system information;
  - For EUTRA cells, multiply $Treselection_{EUTRA}$ by the *sf-Medium* of "Speed dependent ScalingFactor for $Treselection_{EUTRA}$" if broadcasted in system information.

In case scaling is applied to any $Treselection_{RAT}$ parameter, the UE shall round up the result after all scalings to the nearest second.

5.2.4.4 Cells with cell reservations, access restrictions or unsuitable for normal camping

For the highest ranked cell (including serving cell) according to cell reselection criteria specified in clause 5.2.4.6, for the best cell according to absolute priority reselection criteria specified in clause 5.2.4.5, the UE shall check if the access is restricted according to the rules in clause 5.3.1.

If that cell and other cells have to be excluded from the candidate list, as stated in clause 5.3.1, the UE shall not consider these as candidates for cell reselection. This limitation shall be removed when the highest ranked cell changes.

If the highest ranked cell or best cell according to absolute priority reselection rules is an intra-frequency or inter-frequency cell which is not suitable due to one or more of the following reasons:

- this cell belongs to a PLMN which is not indicated as being equivalent to the registered PLMN, or
- this cell is a CAG cell that belongs to a PLMN which is equivalent to the registered PLMN but with no CAG-ID that is present in the UE's allowed CAG list being broadcasted, or
- this cell is not a CAG cell and the CAG-only indication in the UE is set, or
- this cell does not belong to a SNPN that is equal to or indicated as being equivalent to the registered or selected SNPN of the UE in SNPN access mode,

the UE shall not consider this cell and, for operation in licensed spectrum, other cells on the same frequency as candidates for reselection for a maximum of 300 seconds.

For operation with shared spectrum channel access, when the highest ranked cell or best cell is not a candidate for reselection per the previous paragraph, the UE should continue to consider other cells on the same frequency for cell reselection, however if the second highest ranked cell on this frequency is also not suitable due to one or more of the above reasons, the UE may consider this frequency to be the lowest priority for a maximum of 300 seconds.

If the highest ranked cell or best cell according to absolute priority reselection rules is an intra-frequency or inter-frequency cell which is not suitable due to being part of the "list of 5GS forbidden TAs for roaming", the UE shall not consider this cell and other cells on the same frequency as candidates for reselection for a maximum of 300 seconds.

If the highest ranked cell or best cell according to absolute priority reselection rules is an inter-RAT cell which is not suitable due to being part of the "list of forbidden TAs for roaming" or belonging to a PLMN which is not indicated as being equivalent to the registered PLMN, the UE shall not consider this cell and other cells on the same frequency, as candidates for reselection for a maximum of 300 seconds.

If the highest ranked cell or best cell according to absolute priority reselection rules is an intra-frequency or inter-frequency or inter-RAT cell which is not suitable due to its access technology being restricted based on the list of "PLMNs with associated access technology restrictions" (TS 23.122 [9]), the UE shall not consider this cell and other cells of the restricted access technology on the same frequency as candidates for reselection for a maximum of 300 seconds.

If the UE enters into state *any cell selection*, any limitation shall be removed. If the UE is redirected under NR control to a frequency for which the timer is running, the limitation(s) on that frequency shall be removed.

5.2.4.5 NR Inter-frequency and inter-RAT Cell Reselection criteria

If *threshServingLowQ* is broadcast in system information and more than 1 second has elapsed since the UE camped on the current serving cell, cell reselection to a cell on a higher priority NR frequency or inter-RAT frequency than the serving frequency shall be performed if:

- A cell of a higher priority NR or EUTRAN RAT/frequency fulfils Squal > $Thresh_{X, HighQ}$ during a time interval $Treselection_{RAT}$

Otherwise, cell reselection to a cell on a higher priority NR frequency or inter-RAT frequency than the serving frequency shall be performed if:

- A cell of a higher priority RAT/ frequency fulfils Srxlev > $Thresh_{X, HighP}$ during a time interval $Treselection_{RAT}$; and
- More than 1 second has elapsed since the UE camped on the current serving cell.

Cell reselection to a cell on an equal priority NR frequency shall be based on ranking for intra-frequency cell reselection as defined in clause 5.2.4.6.

If *threshServingLowQ* is broadcast in system information and more than 1 second has elapsed since the UE camped on the current serving cell, cell reselection to a cell on a lower priority NR frequency or inter-RAT frequency than the serving frequency shall be performed if:

- The serving cell fulfils Squal < $Thresh_{Serving, LowQ}$ and a cell of a lower priority NR or E-UTRAN RAT/ frequency fulfils Squal > $Thresh_{X, LowQ}$ during a time interval $Treselection_{RAT}$.

Otherwise, cell reselection to a cell on a lower priority NR frequency or inter-RAT frequency than the serving frequency shall be performed if:

- The serving cell fulfils Srxlev < $Thresh_{Serving, LowP}$ and a cell of a lower priority RAT/ frequency fulfils Srxlev > $Thresh_{X, LowP}$ during a time interval $Treselection_{RAT}$; and
- More than 1 second has elapsed since the UE camped on the current serving cell.

Cell reselection to a higher priority RAT/frequency shall take precedence over a lower priority RAT/frequency if multiple cells of different priorities fulfil the cell reselection criteria.

If more than one cell meets the above criteria, the UE shall reselect a cell as follows:

- If the highest-priority frequency is an NR frequency, the highest ranked cell among the cells on the highest priority frequency(ies) meeting the criteria according to clause 5.2.4.6;
- If the highest-priority frequency is from another RAT, the strongest cell among the cells on the highest priority frequency(ies) meeting the criteria of that RAT.

5.2.4.6 Intra-frequency and equal priority inter-frequency Cell Reselection criteria

The cell-ranking criterion R_s for serving cell and R_n for neighbouring cells is defined by:

$R_s = Q_{meas,s} + Q_{hyst} - Qoffset_{temp}$

$R_n = Q_{meas,n} - Qoffset - Qoffset_{temp}$

where:

Q_{meas}	RSRP measurement quantity used in cell reselections.
Qoffset	For intra-frequency: Equals to $Qoffset_{s,n}$, if $Qoffset_{s,n}$ is valid, otherwise this equals to zero. For inter-frequency: Equals to $Qoffset_{s,n}$ plus $Qoffset_{frequency}$, if $Qoffset_{s,n}$ is valid, otherwise this equals to $Qoffset_{frequency}$.
$Qoffset_{temp}$	Offset temporarily applied to a cell as specified in TS 38.331 [3].

The UE shall perform ranking of all cells that fulfil the cell selection criterion S, which is defined in 5.2.3.2.

The cells shall be ranked according to the R criteria specified above by deriving $Q_{meas,n}$ and $Q_{meas,s}$ and calculating the R values using averaged RSRP results.

If *rangeToBestCell* is not configured, the UE shall perform cell reselection to the highest ranked cell. If this cell is found to be not-suitable, the UE shall behave according to clause 5.2.4.4.

If *rangeToBestCell* is configured, then the UE shall perform cell reselection to the cell with the highest number of beams above the threshold (i.e. *absThreshSS-BlocksConsolidation*) among the cells whose R value is within *rangeToBestCell* of the R value of the highest ranked cell. If there are multiple such cells, the UE shall perform cell reselection to the highest ranked cell among them. If this cell is found to be not-suitable, the UE shall behave according to clause 5.2.4.4.

In all cases, the UE shall reselect the new cell, only if the following conditions are met:

- the new cell is better than the serving cell according to the cell reselection criteria specified above during a time interval $Treselection_{RAT}$;
- more than 1 second has elapsed since the UE camped on the current serving cell.

NOTE: If *rangeToBestCell* is configured but *absThreshSS-BlocksConsolidation* is not configured on an NR frequency, the UE considers that there is one beam above the threshold for each cell on that frequency.

5.2.4.7 Cell reselection parameters in system information broadcasts

5.2.4.7.0 General reselection parameters

Cell reselection parameters are broadcast in system information and are read from the serving cell as follows:

absThreshSS-BlocksConsolidation

This specifies the minimum threshold for beams which can be used for selection of the highest ranked cells, if *rangeToBestCell* is configured, and for beams used for derivation of cell measurement quantity. The parameter in *SIB2* applies to the current serving frequency and the parameter in *SIB4* applies to the corresponding inter-frequency.

cellReselectionPriority

This specifies the absolute priority for NR frequency or E-UTRAN frequency.

cellReselectionSubPriority

This specifies the fractional priority value added to cellReselectionPriority for NR frequency or E-UTRAN frequency.

combineRelaxedMeasCondition

This indicates when the UE needs to fulfil both low mobility criterion and not-at-cell-edge criterion to determine whether to relax measurement requirements.

combineRelaxedMeasCondition2

This indicates when an (e)RedCap UE needs to fulfil both stationary criterion and not-at-cell-edge criterion to determine whether to relax measurement requirements.

coverageAreaInfoList

This indicates a list of TN coverage areas to assist skipping TN measurements for NTN UEs in RRC_IDLE and RRC_INACTIVE states.

distanceThresh

This indicates the distance threshold from the serving cell reference location to be used in location-based measurement initiation.

movingReferenceLocation

This indicates the reference location of the serving cell at a time reference, to be used in location-based measurement initiation for NTN Earth-moving cell.

nrofSS-BlocksToAverage

This specifies the number of beams which can be used for selection of the highest ranked cell, if *rangeToBestCell* is configured, and the number of beams used for derivation of cell measurement quantity. The parameter in *SIB2* applies to the current serving frequency and the parameter in *SIB4* applies to the corresponding inter-frequency.

$Qoffset_{s,n}$

This specifies the offset between the two cells.

$Qoffset_{frequency}$

Frequency specific offset for equal priority NR frequencies.

Q_{hyst}

This specifies the hysteresis value for ranking criteria.

$Qoffset_{temp}$

This specifies the additional offset to be used for cell selection and re-selection. It is temporarily used in case the RRC Connection Establishment fails on the cell as specified in TS 38.331 [3].

$Q_{qualmin}$

This specifies the minimum required quality level in the cell in dB.

$Q_{rxlevmin}$

This specifies the minimum required Rx level in the cell in dBm.

$Q_{rxlevminoffsetcell}$

This specifies the cell specific Rx level offset in dB to Qrxlevmin.

$Q_{qualminoffsetcell}$

This specifies the cell specific quality level offset in dB to Qqualmin.

rangeToBestCell

This specifies the R value range which the cells whose R value is within the range can be a candidate for the highest ranked cell. It is configured in SIB2 and used for intra-frequency and equal priority inter-frequency cell reselection and among the cells on the highest priority frequency(ies) for inter-frequency cell reselection within NR.

referenceLocation

This indicates the reference location of the serving cell to be used in location-based measurement initiation for NTN (quasi-)Earth-fixed cell.

$S_{IntraSearchP}$

This specifies the Srxlev threshold (in dB) for intra-frequency measurements.

$S_{IntraSearchQ}$

This specifies the Squal threshold (in dB) for intra-frequency measurements.

$S_{nonIntraSearchP}$

This specifies the Srxlev threshold (in dB) for NR inter-frequency and inter-RAT measurements.

$S_{nonIntraSearchQ}$

This specifies the Squal threshold (in dB) for NR inter-frequency and inter-RAT measurements.

$S_{SearchDeltaP}$

This specifies the threshold (in dB) on Srxlev variation for relaxed measurement.

$S_{SearchDeltaP-Stationary}$

This specifies the threshold (in dB) on Srxlev variation to evaluate stationary criterion for relaxed measurement.

$S_{SearchThresholdP}$

This specifies the Srxlev threshold (in dB) for relaxed measurement.

$S_{SearchThresholdP2}$

This specifies the Srxlev threshold (in dB) to evaluate not-at-cell-edge-criterion for relaxed measurement.

$S_{SearchThresholdQ}$

This specifies the Squal threshold (in dB) for relaxed measurement.

$S_{SearchThresholdQ2}$

This specifies the Squal threshold (in dB) to evaluate not-at-cell-edge-criterion for relaxed measurement.

$Treselection_{RAT}$

This specifies the cell reselection timer value. For each target NR frequency and for each RAT other than NR, a specific value for the cell reselection timer is defined, which is applicable when evaluating reselection within NR or towards other RAT (i.e. $Treselection_{RAT}$ for NR is $Treselection_{NR}$, for E-UTRAN $Treselection_{EUTRA}$).

NOTE: $Treselection_{RAT}$ is not broadcast in system information but used in reselection rules by the UE for each RAT.

$Treselection_{NR}$

This specifies the cell reselection timer value $Treselection_{RAT}$ for NR. The parameter can be set per NR frequency as specified in TS 38.331 [3].

$Treselection_{EUTRA}$

This specifies the cell reselection timer value $Treselection_{RAT}$ for E-UTRAN.

$Thresh_{X, HighP}$

This specifies the Srxlev threshold (in dB) used by the UE when reselecting towards a higher priority RAT/ frequency than the current serving frequency. Each frequency of NR and E-UTRAN might have a specific threshold.

$Thresh_{X, HighQ}$

This specifies the Squal threshold (in dB) used by the UE when reselecting towards a higher priority RAT/ frequency than the current serving frequency. Each frequency of NR and E-UTRAN might have a specific threshold.

$Thresh_{X, LowP}$

This specifies the Srxlev threshold (in dB) used by the UE when reselecting towards a lower priority RAT/ frequency than the current serving frequency. Each frequency of NR and E-UTRAN might have a specific threshold.

$Thresh_{X, LowQ}$

This specifies the Squal threshold (in dB) used by the UE when reselecting towards a lower priority RAT/ frequency than the current serving frequency. Each frequency of NR and E-UTRAN might have a specific threshold.

$Thresh_{Serving, LowP}$

This specifies the Srxlev threshold (in dB) used by the UE on the serving cell when reselecting towards a lower priority RAT/ frequency.

$Thresh_{Serving, LowQ}$

This specifies the Squal threshold (in dB) used by the UE on the serving cell when reselecting towards a lower priority RAT/ frequency.

$T_{SearchDeltaP}$

This specifies the time period over which the Srxlev variation is evaluated for relaxed measurement.

$T_{SearchDeltaP-Stationary}$

This specifies the time period over which the Srxlev variation is evaluated for stationary criterion for relaxed measurement.

t-Service

This indicates the time when an NTN cell is going to stop serving the area where it is currently covering, to be used in time-based measurement initiation.

tn-AreaIdList

This indicates a list of TN area identities associated with each frequency to assist skipping TN measurements for NTN UEs in RRC_IDLE and RRC_INACTIVE states. Each TN area identity in the list identifies a TN coverage area.

5.2.4.7.1 Speed dependent reselection parameters

Speed dependent reselection parameters are broadcast in system information and are read from the serving cell as follows:

T_{CRmax}

This specifies the duration for evaluating allowed amount of cell reselection(s).

N_{CR_M}

This specifies the maximum number of cell reselections to enter Medium-mobility state.

N_{CR_H}

This specifies the maximum number of cell reselections to enter High-mobility state.

$T_{CRmaxHyst}$

This specifies the additional time period before the UE can enter Normal-mobility state.

Speed dependent ScalingFactor for Qhyst

This specifies scaling factor for Qhyst in *sf-High* for High-mobility state and *sf-Medium* for Medium-mobility state.

Speed dependent ScalingFactor for $Treselection_{NR}$

This specifies scaling factor for $Treselection_{NR}$ in *sf-High* for High-mobility state and *sf-Medium* for Medium-mobility state.

Speed dependent ScalingFactor for $Treselection_{EUTRA}$

This specifies scaling factor for $Treselection_{EUTRA}$ in *sf-High* for High-mobility state and *sf-Medium* for Medium-mobility state.

5.2.4.7.2 Slice-based cell reselection parameters

Slice-based cell reselection parameters are broadcast in system information and are read from the serving cell as follows:

nsag-CellReselectionPriority

This specifies the priority for NR frequency when the given NSAG ID is used to set the frequency priority.

nsag-CellReselectionSubPriority

This specifies the fractional priority value added to *nsag-CellReselectionPriority* when the given NSAG ID is used to set the frequency priority.

5.2.4.8 Inter-RAT Cell reselection in RRC_INACTIVE state

For UE in the RRC_INACTIVE state, upon cell reselection to another RAT, UE transitions from RRC_INACTIVE to RRC_IDLE and performs actions as specified in TS 38.331 [3].

5.2.4.9 Relaxed measurement

5.2.4.9.0 Relaxed measurement rules

When the UE is required to perform measurements of intra-frequency cells or NR inter-frequency cells or inter-RAT frequency cells according to the measurement rules in clause 5.2.4.2:

- if *lowMobilityEvaluation* is configured and *cellEdgeEvaluation* is not configured; and
- if the UE has performed normal intra-frequency, NR inter-frequency, or inter-RAT frequency measurements for at least $T_{SearchDeltaP}$ after (re-)selecting a new cell; and
- if the relaxed measurement criterion in clause 5.2.4.9.1 is fulfilled for a period of $T_{SearchDeltaP}$:
  - the UE may choose to perform relaxed measurements for intra-frequency cells, NR inter-frequency cells or inter-RAT frequency cells according to relaxation methods in clauses 4.2.2.9, 4.2.2.10, 4.2.2.11, 4.2C.2.7 and 4.2C.2.8 in TS 38.133 [8];
- if *cellEdgeEvaluation* is configured and *lowMobilityEvaluation* is not configured; and
- if the relaxed measurement criterion in clause 5.2.4.9.2 is fulfilled:
  - the UE may choose to perform relaxed measurements for intra-frequency cells according to relaxation methods in clauses 4.2.2.9 and 4.2C.2.7 in TS 38.133 [8];
  - if the serving cell fulfils Srxlev $\leq$ $S_{nonIntraSearchP}$ or Squal $\leq$ $S_{nonIntraSearchQ}$:
    - the UE may choose to perform relaxed measurements for NR inter-frequency cells or inter-RAT frequency cells according to relaxation methods in clauses 4.2.2.10, 4.2.2.11 and 4.2C.2.8 in TS 38.133 [8];
- if both *lowMobilityEvaluation* and *cellEdgeEvaluation* are configured:
  - if the UE has performed normal intra-frequency, NR inter-frequency, or inter-RAT frequency measurements for at least $T_{SearchDeltaP}$ after (re-)selecting a new cell; and
  - if the relaxed measurement criterion in clause 5.2.4.9.1 is fulfilled for a period of $T_{SearchDeltaP}$; and
  - if the relaxed measurement criterion in clause 5.2.4.9.2 is fulfilled:
    - the UE may choose to perform relaxed measurements for NR intra-frequency cells, inter-frequency cells or inter-RAT frequency cells according to relaxation methods in clauses 4.2.2.9, 4.2.2.10, 4.2.2.11, 4.2C.2.7 and 4.2C.2.8 in TS 38.133 [8];
  - else:
    - if the UE has performed normal intra-frequency, NR inter-frequency, or inter-RAT frequency measurements for at least $T_{SearchDeltaP}$ after (re-)selecting a new cell, and the relaxed measurement criterion in clause 5.2.4.9.1 is fulfilled for a period of $T_{SearchDeltaP}$; or,
    - if the relaxed measurement criterion in clause 5.2.4.9.2 is fulfilled:
      - if *combineRelaxedMeasCondition* is not configured:
        - the UE may choose to perform relaxed measurements for NR intra-frequency cells, inter-frequency cells or inter-RAT frequency cells according to relaxation methods in clauses 4.2.2.9, 4.2.2.10, 4.2.2.11, 4.2C.2.7 and 4.2C.2.8 in TS 38.133 [8];

- if the UE is an (e)RedCap UE; and
- if *stationaryMobilityEvaluation* is configured and *cellEdgeEvaluationWhileStationary* is not configured; and
- if the UE has performed normal intra-frequency, NR inter-frequency, or inter-RAT frequency measurements for at least $T_{SearchDeltaP-Stationary}$ after (re-)selecting a new cell; and
- if the relaxed measurement criterion in clause 5.2.4.9.3 is fulfilled for a period of $T_{SearchDeltaP-Stationary}$:
  - the UE may choose to perform relaxed measurements for intra-frequency cells, NR inter-frequency cells, or inter-RAT frequency cells according to relaxation methods in clauses 4.2B.2.9, 4.2B.2.10, and 4.2B.2.11 in TS 38.133 [8];
- if the UE is an (e)RedCap UE; and
- if both *stationaryMobilityEvaluation* and *cellEdgeEvaluationWhileStationary* are configured:
  - if the UE has performed normal intra-frequency, NR inter-frequency, or inter-RAT frequency measurements for at least $T_{SearchDeltaP-Stationary}$ after (re-)selecting a new cell; and
  - if the relaxed measurement criterion in clause 5.2.4.9.4 is fulfilled:
    - the UE may choose to perform relaxed measurements for intra-frequency cells, NR inter-frequency cells, or inter-RAT frequency cells according to relaxation methods in clauses 4.2B.2.9, 4.2B.2.10, and 4.2B.2.11 in TS 38.133 [8];
  - else:
    - if *combineRelaxedMeasCondition2* is not configured:
      - if the UE has performed normal intra-frequency, NR inter-frequency, or inter-RAT frequency measurements for at least $T_{SearchDeltaP-Stationary}$ after (re-)selecting a new cell; and
      - if the relaxed measurement criterion in clause 5.2.4.9.3 is fulfilled for a period of $T_{SearchDeltaP-Stationary}$:
        - the UE may choose to perform relaxed measurements for intra-frequency cells, NR inter-frequency cells, or inter-RAT frequency cells according to relaxation methods in clauses 4.2B.2.9, 4.2B.2.10, and 4.2B.2.11 in TS 38.133 [8];

NOTE 1: It is up to UE implementation when to start performing relaxed measurements in RRC Idle/Inactive if multiple methods are configured.

NOTE 2: It is up to UE implementation which relaxation method to perform based on the "allowed" cases as specified in TS 38.133 [8] for RRC Idle/Inactive if multiple methods are configured.

The above relaxed measurements and no measurement are not applicable for frequencies that are included in *VarMeasIdleConfig*, if configured and for which the UE supports dual connectivity or carrier aggregation between those frequencies and the frequency of the current serving cell.

5.2.4.9.1 Relaxed measurement criterion for UE with low mobility

The relaxed measurement criterion for UE with low mobility is fulfilled when:

- $(Srxlev_{Ref} - Srxlev) < S_{SearchDeltaP}$,

Where:

- Srxlev = current Srxlev value of the serving cell (dB).
- $Srxlev_{Ref}$ = reference Srxlev value of the serving cell (dB), set as follows:
  - After selecting or reselecting a new cell, or
  - If $(Srxlev - Srxlev_{Ref}) > 0$, or
  - If the relaxed measurement criterion has not been met for $T_{SearchDeltaP}$:

- The UE shall set the value of $Srxlev_{Ref}$ to the current Srxlev value of the serving cell.

5.2.4.9.2 Relaxed measurement criterion for UE not at cell edge

The relaxed measurement criterion for UE not at cell edge is fulfilled when:

- Srxlev > $S_{SearchThresholdP}$, and,
- Squal > $S_{SearchThresholdQ}$, if $S_{SearchThresholdQ}$ is configured,

Where:

- Srxlev = current Srxlev value of the serving cell (dB).
- Squal = current Squal value of the serving cell (dB).

5.2.4.9.3 Relaxed measurement criterion for a stationary (e)RedCap UE

The relaxed measurement criterion for a stationary (e)RedCap UE is fulfilled when:

- ($Srxlev_{RefStationary}$ – Srxlev) < $S_{SearchDeltaP-Stationary}$,

Where:

- Srxlev = current Srxlev value of the serving cell (dB).
- $Srxlev_{RefStationary}$ = reference Srxlev value of the serving cell (dB), set as follows:
  - After selecting or reselecting a new cell, or
  - If (Srxlev - $Srxlev_{RefStationary}$) > 0, or
  - If the relaxed measurement criterion has not been met for $T_{SearchDeltaP-Stationary}$:
    - The UE shall set the value of $Srxlev_{RefStationary}$ to the current Srxlev value of the serving cell.

5.2.4.9.4 Relaxed measurement criterion for a stationary (e)RedCap UE not at cell edge

The relaxed measurement criterion for a stationary (e)RedCap UE not at cell edge is fulfilled when:

- the relaxed measurement criterion in clause 5.2.4.9.3 is fulfilled for a period of $T_{SearchDeltaP-Stationary}$, and,
- Srxlev > $S_{SearchThresholdP2}$, and,
- Squal > $S_{SearchThresholdQ2}$, if $S_{SearchThresholdQ2}$ is configured.

Where:

- Srxlev = current Srxlev value of the serving cell (dB).
- Squal = current Squal value of the serving cell (dB).

5.2.4.10 Cell reselection with CAG cells

In addition to normal cell reselection, a UE may optionally use an autonomous search function to detect CAG cells on serving and non-serving frequencies. However UE shall follow the cell reselection criteria based on dedicated frequency priorities and only follow the autonomous cell search result if the result fulfils also the existing cell reselection criteria based on dedicated frequency priorities.

NOTE: Mobile-IAB cell reselection priority handling as specified in clause 5.2.4.1 is applicable for a mobile-IAB cell irrespective of whether this cell is a CAG cell or not.

5.2.4.11 Reselection priorities for slice-based cell reselection

The UE derives reselection priorities for slice-based cell reselection by using:

- NAS provided NSAG information, only for NSAG(s) associated with the network slice(s) provided by NAS for cell reselection (see TS 23.501 [10], TS 24.501 [14]),

- *sliceInfoList* and/or *sliceInfoListDedicated* per frequency with *nsag-CellReselectionPriority* per NSAG, if provided in system information and/or dedicated signalling (see TS 38.331 [3]),

- *cellReselectionPriority* per frequency provided in system information and/or dedicated signalling (see TS 38.331 [3]).

The UE considers an NR frequency to support all slices of an NSAG if

- the nsag-ID and TA of the NSAG indicated for the NR frequency (see TS 38.331[3]) are included in the NSAG information provided by NAS. If *FreqPriorityListDedicatedSlicing* is configured, UE only considers the NSAG-frequency pairs indicated in *FreqPriorityListDedicatedSlicing* for slice-based cell reselection.

The UE considers a cell on an NR frequency to support all slices of an NSAG if

- the nsag-ID and TA of the NSAG indicated for the NR frequency in dedicated signalling but not in *SIB16* (see TS 38.331 [3]) are included in the NSAG information provided by NAS; or

- the nsag-ID and TA of the NSAG indicated for the NR frequency in *SIB16* (see TS 38.331 [3]) are included in the NSAG information provided by NAS; and

  - the cell is either listed in the *sliceAllowedCellListNR* (if provided in the *sliceInfoList*) or the cell is not listed in the *sliceExcludedCellListNR* (if provided in the *sliceInfoList*); or

  - Neither *sliceAllowedCellListNR* nor *sliceExcludedCellListNR* is configured in the *sliceInfoList*.

The UE shall derive reselection priorities for slice-based cell reselection according to the following rules:

- Frequencies that support at least one prioritized NSAG received from NAS have higher reselection priority than frequencies that support none of the NSAG(s) received from NAS.

- Frequencies that support at least one NSAG provided by NAS are prioritised in the order of the NAS-provided priority for the NSAG with highest priority supported on the frequency.

- Among the frequencies (one or multiple) that support the highest prioritised NSAG(s) with the same NAS-provided priorities, the frequencies are prioritized in the order of their highest *nsag-CellReselectionPriority* given for these NSAG(s). If no *nsag-CellReselectionPriority* is given for a NSAG at a frequency, the lowest priority value is used (i.e, lower than any of the network configured values for these frequencies).

- Frequencies that support none of the NSAG(s) provided by NAS are prioritized in the order of their *cellReselectionPriority*.

For a UE performing slice-based cell reselection, if the highest ranked cell or best cell in a frequency fulfils the inter-freqeuency cell reselection criteria (see clause 5.2.4.5) based on reselection priority for the frequency and NSAG derived according to this clause or fulfils intra-frequency and equal priority inter-frequency cell reselection criteria (see clause 5.2.4.6), but this cell does not support the NSAG according to this clause:

- if this cell supports any other NSAG(s) according to this clause, the UE shall re-derive a reselection priority for the frequency by considering the NSAG(s) supported by this cell (rather than those of the corresponding NR frequency);

- Otherwise, the UE shall re-derive a reselection priority for the frequency as if none of the NSAG(s) provided by NAS is supported.

This re-derived reselection priority is used for a maximum of 300 seconds, or until new network slice(s) and/or NSAG information are received from NAS. UE shall ensure the cell reselection criteria above are fulfilled based on the newly derived priorities.

5.2.4.12 Relaxed measurement and measurement offloading for UE supporting LP-WUS

In multi-beam operations, measured cell RX level value and measured cell quality value of the serving cell based on LR is up to UE implementation.

5.2.4.12.1 Relaxed measurement rules

UE supporting LP-WUS may choose to perform relaxed serving cell and further relaxed neighbouring cell measurements on MR according to requirements specified in TS 38.133 [8] if the relaxed measurement criterion in clause 5.2.4.12.2 is fulfilled.

5.2.4.12.2 Relaxed measurement criterion

The relaxed measurement criterion for serving cell and neighbouring cell measurement relaxation on MR is fulfilled when:

- Srxlev > $S_{LP_WUS_RelaxThresholdP_MR}$, and,
- $Q_{rxlevmeas_lr}$ > $Q_{LP_WUS_RelaxThresholdP_LR}$, if $Q_{LP_WUS_RelaxThresholdP_LR}$ is configured, and,
- Squal > $S_{LP_WUS_RelaxThresholdQ_MR}$, if $S_{LP_WUS_RelaxThresholdQ_MR}$ is configured, and
- $Q_{qualmeas_lr}$ > $Q_{LP_WUS_RelaxThresholdQ_LR}$, if $Q_{LP_WUS_RelaxThresholdQ_LR}$ is configured,

Where:

- Srxlev = current Srxlev value of the serving cell (dB).
- Squal = current Squal value of the serving cell (dB).
- $Q_{rxlevmeas_lr}$ = current measured cell RX level value of the serving cell based on LR (RSRP).
- $Q_{qualmeas_lr}$ = current measured cell quality value of the serving cell based on LR (RSRQ).
- $S_{LP_WUS_RelaxThresholdP_MR}$: The Srxlev threshold for the criterion for serving cell and neighbouring cell measurement relaxation based on MR.
- $S_{LP_WUS_RelaxThresholdQ_MR}$: The Squal threshold for the criterion for serving cell and neighbouring cell measurement relaxation based on MR.
- $Q_{LP_WUS_RelaxThresholdP_LR}$: The cell RX level threshold for the criterion for serving cell and neighbouring cell measurement relaxation based on LR.
- $Q_{LP_WUS_RelaxThresholdQ_LR}$: The cell quality threshold for the criterion for serving cell and neighbouring cell measurement relaxation based on LR.

These thresholds can be configured separately for LR measurements based on LP-SS and LR measurements based on SSB if a cell supports both measurement types as specified in TS 38.331 [3]. $S_{LP_WUS_RelaxThresholdP_MR}$ is the parameter $S_{SearchThresholdP3}$ or $S_{SearchThresholdP4}$ in TS 38.331 [3]. $S_{LP_WUS_RelaxThresholdQ_MR}$ is the parameter $S_{SearchThresholdQ3}$ or $S_{SearchThresholdQ4}$ in TS 38.331 [3]. $Q_{LP_WUS_RelaxThresholdP_LR}$ is the parameter $S_{RSRPThresholdLR}$ or $S_{RSRPThresholdLR2}$ in TS 38.331 [3]. $Q_{LP_WUS_RelaxThresholdQ_LR}$ is the parameter $S_{RSRQThresholdLR}$ or $S_{RSRQThresholdLR2}$ in TS 38.331 [3]. If UE supports both measurement types, it is up to UE implementation to choose LR measurements based on LP-SS or based on SSB for the determination of the relaxed measurement criterion for serving cell and neighbouring cell measurement relaxation on MR.

5.2.4.12.3 Serving cell measurement offloading rules

UE supporting LP-WUS may choose to perform serving cell measurement offloading (i.e., serving cell measurement is fully offloaded to LR and no serving cell measurement via MR is required) according to requirements specified in TS 38.133 [8] if the entry condition for serving cell measurement offloading in clause 5.2.4.12.4 is fulfilled. UE supporting LP-WUS is required to stop serving cell measurement offloading according to requirements specified in TS 38.133 [8] if the exit condition for serving cell measurement offloading in clause 5.2.4.12.4 is fulfilled.

5.2.4.12.4 Serving cell measurement offloading criterion

The entry condition for serving cell measurement offloading is fulfilled when:

- Srxlev > $S_{LP_WUS_offloadingEntryThresholdP_MR}$, and,

- $Q_{rxlevmeas_lr} > Q_{LP_WUS_offloadingEntryThresholdP_LR}$, if $Q_{LP_WUS_offloadingxEntryThresholdP_LR}$ is configured, and,
- Squal $> S_{LP_WUS_offloadingEntryThresholdQ_MR}$, if $S_{LP_WUS_offloadingEntryThresholdQ_MR}$ is configured, and
- $Q_{qualmeas_lr} > Q_{LP_WUS_offloadingEntryThresholdQ_LR}$, if $Q_{LP_WUS_offloadingEntryThresholdQ_LR}$ is configured.

The exit condition for serving cell measurement offloading is fulfilled when:

- $Q_{rxlevmeas_lr} < Q_{LP_WUS_offloadingExitThresholdP_LR}$, or,
- $Q_{qualmeas_lr} < Q_{LP_WUS_offloadingExitThresholdQ_LR}$, if $Q_{LP_WUS_offloadingExitThresholdQ_LR}$ is configured.

Where:

- Srxlev = current Srxlev value of the serving cell (dB).
- Squal = current Squal value of the serving cell (dB).
- $Q_{rxlevmeas_lr}$= current measured cell RX level value of the serving cell based on LR (RSRP).
- $Q_{qualmeas_lr}$ = current measured cell quality value of the serving cell based on LR (RSRQ).
- $S_{LP_WUS_offloadingEntryThresholdP_MR}$: The Srxlev threshold for entry condition of serving cell measurement offloading based on MR.
- $S_{LP_WUS_offloadingEntryThresholdQ_MR}$: The Squal threshold for entry condition of serving cell measurement offloading based on MR.
- $Q_{LP_WUS_offloadingEntryThresholdP_LR}$: The cell RX level threshold for entry condition of serving cell measurement offloading based on LR.
- $Q_{LP_WUS_offloadingEntryThresholdQ_LR}$: The cell quality threshold for entry condition of serving cell measurement offloading based on LR.
- $Q_{LP_WUS_offloadingExitThresholdP_LR}$: The cell RX level threshold for exit condition of serving cell measurement offloading based on LR.
- $Q_{LP_WUS_offloadingExitThresholdQ_LR}$: The cell quality threshold for exit condition of serving cell measurement offloading based on LR.

These thresholds can be configured separately for LR measurements based on LP-SS and LR measurements based on SSB if a cell supports both measurement types as specified in TS 38.331 [3]. $S_{LP_WUS_offloadingEntryThresholdP_MR}$ is the parameter $S_{SearchThresholdP5}$ or $S_{SearchThresholdP6}$ in TS 38.331 [3]. $S_{LP_WUS_offloadingEntryThresholdQ_MR}$ is the parameter $S_{SearchThresholdQ5}$ or $S_{SearchThresholdQ6}$ in TS 38.331 [3]. $Q_{LP_WUS_offloadingEntryThresholdP_LR}$ is the parameter $S_{RSRPThresholdLR3}$ or $S_{RSRPThresholdLR4}$ in TS 38.331 [3]. $Q_{LP_WUS_offloadingEntryThresholdQ_LR}$ is the parameter $S_{RSRQThresholdLR3}$ or $S_{RSRQThresholdLR4}$ in TS 38.331 [3]. $Q_{LP_WUS_offloadingExitThresholdP_LR}$ is the parameter $S_{RSRPThresholdLR5}$ or $S_{RSRPThresholdLR6}$ in TS 38.331 [3]. $Q_{LP_WUS_offloadingExitThresholdQ_LR}$ is the parameter $S_{RSRQThresholdLR5}$ or $S_{RSRQThresholdLR6}$ in TS 38.331 [3]. If UE supports both measurement types, it is up to UE implementation to choose LR measurements based on LP-SS or based on SSB for the determination of the entry/exit conditions of serving cell measurement offloading.

5.2.5 Camped Normally state

This state is applicable for RRC_IDLE and RRC_INACTIVE state.

When camped normally, the UE shall perform the following tasks:

- monitor the paging channel of the cell as specified in clause 7 according to information broadcast in *SIB1*;
- monitor Short Messages transmitted with P-RNTI over DCI as specified in clause 6.5 in TS 38.331 [3];
- monitor relevant System Information as specified in TS 38.331 [3];
- perform necessary measurements for the cell reselection evaluation procedure;
- execute the cell reselection evaluation process on the following occasions/triggers:

1) UE internal triggers, so as to meet performance as specified in TS 38.133 [8];

2) When information on the BCCH used for the cell reselection evaluation procedure has been modified.

3) When the network slice(s) and/or NSAG information received from NAS changes.

5.2.6 Selection of cell at transition to RRC_IDLE or RRC_INACTIVE state

At reception of *RRCRelease* message to transition the UE to RRC_IDLE or RRC_INACTIVE, UE shall attempt to camp on a suitable cell according to *redirectedCarrierInfo* if included in the *RRCRelease* message. If the UE cannot find a suitable cell, the UE is allowed to camp on any suitable cell of the indicated RAT. If the *RRCRelease* message does not contain the *redirectedCarrierInfo*, UE shall attempt to select a suitable cell on an NR carrier. If no suitable cell is found according to the above, the UE shall perform cell selection using stored information in order to find a suitable cell to camp on.

When returning to RRC_IDLE state after UE moved to RRC_CONNECTED state from *camped on any cell* state, UE shall attempt to camp on an acceptable cell according to *redirectedCarrierInfo*, if included in the *RRCRelease* message. If the UE cannot find an acceptable cell, the UE is allowed to camp on any acceptable cell of the indicated RAT. If the *RRCRelease* message does not contain *redirectedCarrierInfo* UE shall attempt to select an acceptable cell on an NR frequency. If no acceptable cell is found according to the above, the UE not in SNPN Access Mode shall continue to search for an acceptable cell of any PLMN in state *any cell selection*. If no acceptable cell is found according to the above, the UE in SNPN access mode shall continue to search for an acceptable cell of any SNPN in state *any cell selection*.

5.2.7 Any Cell Selection state

This state is applicable for RRC_IDLE and RRC_INACTIVE state. In this state, the UE shall perform cell selection process to find a suitable cell. If the cell selection process fails to find a suitable cell after a complete scan of all RATs and all frequency bands supported by the UE, the UE not in SNPN Access Mode shall attempt to find an acceptable cell of any PLMN to camp on, trying all RATs that are supported by the UE and searching first for a high-quality cell, as defined in clause 5.1.1.2. If the cell selection process fails to find a suitable cell after a complete scan of all frequency bands supported by the UE, the UE in SNPN access mode shall attempt to find an acceptable cell of any SNPN to camp on.

The UE, which is not camped on any cell, shall stay in this state.

5.2.8 Camped on Any Cell state

This state is only applicable for RRC_IDLE state. In this state, the UE shall perform the following tasks:

- monitor Short Messages transmitted with P-RNTI over DCI as specified in clause 6.5 in TS 38.331 [3];
- monitor relevant System Information as specified in TS 38.331 [3];
- perform necessary measurements for the cell reselection evaluation procedure;
- execute the cell reselection evaluation process on the following occasions/triggers:
  1) UE internal triggers, so as to meet performance as specified in TS 38.133 [8];
  2) When information on the BCCH used for the cell reselection evaluation procedure has been modified.
- regularly attempt to find a suitable cell trying all frequencies of all RATs that are supported by the UE. If a suitable cell is found, UE shall move to *camped normally* state.
- if the UE supports voice services, the UE is not in SNPN access mode, and the current cell does not support IMS emergency calls as indicated by the field *ims-EmergencySupport* in SIB1 as specified in TS 38.331 [3], the UE shall perform cell selection/reselection to an acceptable cell that supports emergency calls in any supported RAT regardless of priorities provided in system information from current cell, if no suitable cell is found.
- if the UE supports voice services, the UE is in SNPN access mode, and the current cell does not support IMS emergency calls for any SNPN(s) as indicated by the field *imsEmergencySupportForSNPN* in SIB1 as

specified in TS 38.331 [3], the UE shall perform cell selection/reselection to an acceptable cell of any available SNPN that supports emergency calls, if no suitable cell is found.

5.3 Cell Reservations and Access Restrictions

5.3.0 Introduction

There are two mechanisms which allow an operator to impose cell reservations or access restrictions. The first mechanism uses indication of cell status and special reservations for control of cell selection and reselection procedures. The second mechanism, referred to as Unified Access Control as specified in TS 38.331 [3], shall allow preventing selected access categories or access identities from sending initial access messages for load control reasons.

Unified Access Control does not apply to IAB-MTs and NCR-MTs.

5.3.1 Cell status and cell reservations

Cell status and cell reservations are indicated in the *MIB* or *SIB1* message as specified in TS 38.331 [3] by means of the following fields:

- *cellBarred* (IE type: "barred" or "not barred")
  Indicated in *MIB* message. In case of multiple PLMNs or NPNs indicated in *SIB1*, this field is common for all PLMNs and NPNs. This field is ignored by UEs supporting NTN for NTN access, or by UEs supporting ATG for ATG access.

- *cellBarredATG* (IE type: "barred" or "not barred")
  Indicated in *SIB1* message. In case of multiple PLMNs or NPNs indicated in *SIB1*, this field is common for all PLMNs and NPNs. This field is only applicable to ATG UEs.

- *cellBarred2RxXR* (IE type: "barred")
  Indicated in *SIB1* message. In case of multiple PLMNs or NPNs indicated in *SIB1*, this field is common for all PLMNs and NPNs. This field is only applicable to 2Rx XR UEs.

- *cellBarred-eRedCap1Rx* (IE type: "barred" or "not barred")
  Indicated in *SIB1* message. In case of multiple PLMNs or NPNs indicated in *SIB1*, this field is common for all PLMNs and NPNs. This field is only applicable to eRedCap UEs.

- *cellBarred-eRedCap2Rx* (IE type: "barred" or "not barred")
  Indicated in *SIB1* message. In case of multiple PLMNs or NPNs indicated in *SIB1*, this field is common for all PLMNs and NPNs. This field is only applicable to eRedCap UEs.

- *cellBarredFixedVSAT* (IE type: "barred" or "not barred")
  Indicated in *SIB1* message. In case of multiple PLMNs indicated in *SIB1*, this field is common for all PLMNs. This field is only applicable to VSAT UEs using NTN access.

- *cellBarredMobileVSAT* (IE type: "barred" or "not barred")
  Indicated in *SIB1* message. In case of multiple PLMNs indicated in *SIB1*, this field is common for all PLMNs. This field is only applicable to VSAT UEs using NTN access.

- *cellBarredNES* (IE type: "not barred")
  Indicated in *SIB1* message. In case of multiple PLMNs or NPNs indicated in *SIB1*, this field is common for all PLMNs and NPNs. This field is only applicable to UEs indicating any of the values in *nes-CellDTX-DRX* as specified in TS 38.306 [24].

- *cellBarredNTN* (IE type: "barred" or "not barred")
  Indicated in *SIB1* message. In case of multiple PLMNs indicated in *SIB1*, this field is common for all PLMNs. This field is ignored if the UE does not support NTN access.

- *cellBarredRedCap1Rx* (IE type: "barred" or "not barred")
  Indicated in *SIB1* message. In case of multiple PLMNs or NPNs indicated in *SIB1*, this field is common for all PLMNs and NPNs. This field is only applicable to RedCap UEs.

- *cellBarredRedCap2Rx* (IE type: "barred" or "not barred")
  Indicated in *SIB1* message. In case of multiple PLMNs or NPNs indicated in *SIB1*, this field is common for all PLMNs and NPNs. This field is only applicable to RedCap UEs.

- *cellReservedForOperatorUse* (IE type: "reserved" or "not reserved")
  Indicated in *SIB1* message. In case of multiple PLMNs or NPNs indicated in *SIB1*, this field is specified per PLMN or per SNPN.

- *cellReservedForOtherUse* (IE type: "true")
  Indicated in *SIB1* message. In case of multiple PLMNs indicated in *SIB1*, this field is common for all PLMNs.

- *cellReservedForFutureUse* (IE type: "true")
  Indicated in *SIB1* message. In case of multiple PLMNs or NPNs indicated in *SIB1*, this field is common for all PLMNs and NPNs.

NOTE 0: IAB-MT ignores the *cellBarred*, *cellReservedForOperatorUse*, *cellReservedForFutureUse*, and *intraFreqReselection* (i.e. treats *intraFreqReselection* as if it was set to *allowed*) as defined in TS 38.331 [3]. IAB-MT also ignores *cellReservedForOtherUse* for cell barring determination (i.e. NPN capable IAB-MT considers *cellReservedForOtherUse* for determination of an NPN-only cell) as defined in TS 38.331 [3].

NOTE 0a: NCR-MT ignores the *cellBarred*, *cellReservedForOperatorUse*, *cellReservedForFutureUse*, and *intraFreqReselection* (i.e. treats *intraFreqReselection* as if it was set to *allowed*) as defined in TS 38.331 [3]. NCR-MT also ignores *cellReservedForOtherUse* for cell barring determination (i.e. NPN capable NCR-MT considers *cellReservedForOtherUse* for determination of an NPN-only cell) as defined in TS 38.331 [3].

- *halfDuplexRedCapAllowed* (IE type: "true")
  Indicated in *SIB1* message. In case of multiple PLMNs or NPNs indicated in *SIB1*, this field is common for all PLMNs and NPNs. This field is only applicable to (e)RedCap UEs.

- *iab-Support* (IE type: "true")
  Indicated in *SIB1* message. In case of multiple PLMNs or NPNs indicated in *SIB1*, this field is specified per PLMN or per SNPN. This field is only applicable to IAB-MT.

- *ncr-Support* (IE type: "true")
  Indicated in *SIB1* message. In case of multiple PLMNs or NPNs indicated in *SIB1*, this field is common for all PLMNs and NPNs. This field is only applicable to NCR-MT.

- *mobileIAB-Support* (IE type: "true")
  Indicated in *SIB1* message. In case of multiple PLMNs or NPNs indicated in *SIB1*, this field is specified per PLMN or per SNPN. This field is only applicable to mobile IAB-MT.

When cell status is indicated as "not barred" and "not reserved" for operator use and not "true" for other use and not "true" for future use,

- UEs shall treat this cell as candidate during the cell selection and cell reselection procedures.

When cell broadcasts any CAG-IDs or NIDs and the cell status is indicated as "not barred" and "not reserved" for operator use and "true" for other use, and not "true" for future use:

- All NPN-capable UEs shall treat this cell as candidate during the cell selection and cell reselection procedures, other UEs shall treat this cell as if cell status is "barred".

When cell status is indicated as "true" for other use, and either cell does not broadcast any CAG-IDs or NIDs or does not broadcast any CAG-IDs and the UE is not operating in SNPN Access Mode,

- The UE shall treat this cell as if cell status is "barred".

When cell status is indicated as "true" for future use,

- The UE shall treat this cell as if cell status is "barred".

When *cellBarredNES* is absent and *cellBarred* is set to "barred",

- The UE indicating any of the values in *nes-CellDTX-DRX* shall treat this cell as if cell status is "barred".

When *cellBarredNTN* is not broadcast in this cell,

- For NTN access, the UE shall treat this cell as if cell status is "barred".

When *halfDuplexRedCapAllowed* is not broadcast in this cell,

- The (e)RedCap UE only capable of operating in half-duplex for FDD shall treat this cell as if cell status is "barred".

When *cellBarredATG* is not broadcast in this cell,

- For ATG access, the UE shall treat this cell as if cell status is "barred".

When *cellBarredFixedVSAT* is not broadcast in this cell,

- For NTN access, the fixed VSAT UE shall treat this cell as if cell status is "barred".

When *cellBarredMobileVSAT* is not broadcast in this cell,

- For NTN access, the mobile VSAT UE shall treat this cell as if cell status is "barred".

When *cellBarred2RxXR* is broadcast in this cell,

- The 2Rx XR UE shall treat this cell as if cell status is "barred".

When *intraFreqReselectionRedCap* is not broadcast in this cell,

- The RedCap UE shall treat this cell as if cell status is "barred".

When *intraFreqReselection-eRedCap* is not broadcast in this cell,

- The eRedCap UE shall treat this cell as if cell status is "barred".

When *cellBarredRedCap1Rx* is set to "barred" and *barringExemptEmergencyCall* is present in *SIB1*, if the cell will not be treated as barred by the UE for any reason other than the *cellBarredRedCap1Rx* being set to "barred" (see TS 38.331 [3]), and cell selection criteria are fulfilled as defined in clause 5.2.3,

- The RedCap UE that supports barring exemption for emergency call (see TS 38.306 [24]) and only 1Rx branch shall treat this cell as an acceptable cell.

When *cellBarredRedCap2Rx* is set to "barred" and *barringExemptEmergencyCall* is present in *SIB1*, if the cell will not be treated as barred by the UE for any reason other than the *cellBarredRedCap2Rx* being set to "barred" (see TS 38.331 [3]), and cell selection criteria are fulfilled as defined in clause 5.2.3,

- The RedCap UE that supports barring exemption for emergency call (see TS 38.306 [24]) and 2Rx branches shall treat this cell as an acceptable cell.

When *cellBarred-eRedCap1Rx* is set to "barred" and *barringExemptEmergencyCall* is present in *SIB1*, if the cell will not be treated as barred by the UE for any reason other than the *cellBarred-eRedCap1Rx* being set to "barred" (see TS 38.331 [3]), and cell selection criteria are fulfilled as defined in clause 5.2.3,

- The eRedCap UE that supports barring exemption for emergency call (see TS 38.306 [24]) and only 1Rx branch shall treat this cell as an acceptable cell.

When *cellBarred-eRedCap2Rx* is set to "barred" and *barringExemptEmergencyCall* is present in *SIB1*, if the cell will not be treated as barred by the UE for any reason other than the *cellBarred-eRedCap2Rx* being set to "barred" (see TS 38.331 [3]), and cell selection criteria are fulfilled as defined in clause 5.2.3,

- The eRedCap UE that supports barring exemption for emergency call (see TS 38.306 [24]) and 2Rx branches shall treat this cell as an acceptable cell.

When *cellBarred2RxXR* and *barringExemptEmergencyCall* are both present in *SIB1*, if the cell will not be treated as barred by the UE for any reason other than the *cellBarred2RxXR* being present in *SIB1* (see TS 38.331 [3]), and cell selection criteria are fulfilled as defined in clause 5.2.3,

- The 2Rx XR UE that supports barring exemption for emergency call (see TS 38.306 [24]) shall treat this cell as an acceptable cell.

When cell status is indicated as "not barred" and "reserved" for operator use for any PLMN/SNPN and not "true" for other use and not "true" for future use,

- UEs assigned to Access Identity 11 or 15 operating in their HPLMN/EHPLMN shall treat this cell as candidate during the cell selection and reselection procedures if the field *cellReservedForOperatorUse* for that PLMN set to "reserved".
- UEs assigned to Access Identity 11 or 15 shall treat this cell as candidate during the cell selection and reselection procedures if the field *cellReservedForOperatorUse* for selected/registered SNPN is set to "reserved".
- UEs assigned to an Access Identity 0, 1, 2 and 12 to 14 shall behave as if the cell status is "barred" in case the cell is "reserved for operator use" for the registered PLMN/SNPN or the selected PLMN/SNPN.
- UEs assigned to Access Identity 3 shall behave as if the cell status is "barred" in case the cell is "reserved for operator use" for the registered PLMN or the selected PLMN.

NOTE 1: Access Identities 11, 15 are only valid for use in the HPLMN/ EHPLMN and registered/selected SNPN; Access Identities 12, 13, 14 are only valid for use in the home country and registered/selected SNPN as specified in TS 22.261 [12].

NOTE 1a: Access Identity 3 is only valid for PLMNs that indicate to potential Disaster Inbound Roamers that the UEs can access the PLMN as specified in TS 22.261 [12].

When cell status "barred" is indicated or to be treated as if the cell status is "barred",

- The UE is not permitted to select/reselect this cell, not even for emergency calls;
- The UE shall select another cell according to the following rule:
- If the cell is to be treated as if the cell status is "barred" due to being unable to acquire the *MIB*:
  - the UE may exclude the barred cell as a candidate for cell selection/reselection for up to 300 seconds.
  - the UE may select another cell on the same frequency if the selection criteria are fulfilled.
- else:
  - If the UE is a RedCap UE, the UE shall acquire *SIB1* and, in the remainder of this procedure, consider '*intraFreqReselection* in MIB' to be '*intraFreqReselectionRedCap* in *SIB1*', if available; or,
  - If the UE is an eRedCap UE, the UE shall acquire *SIB1* and, in the remainder of this procedure, consider '*intraFreqReselection* in MIB' to be '*intraFreqReselection-eRedCap* in *SIB1*', if available; or,
  - If the UE is a 2Rx XR UE, the UE shall acquire *SIB1* and, in the remainder of this procedure, consider '*intraFreqReselection* in MIB' to be '*intraFreqReselection2RxXR* in *SIB1*', if available:
    - If the cell is to be treated as if the cell status is "barred" due to being unable to acquire the *SIB1*:
      - the UE may exclude the barred cell as a candidate for cell selection/reselection for up to 300 seconds.
      - the UE may select another cell on the same frequency if the selection criteria are fulfilled.
    - If the cell status "barred" is indicated in *MIB* but the UE is unable to acquire the *SIB1*; or
    - If the UE is a RedCap UE and *intraFreqReselectionRedCap* in *SIB1* is not available; or
    - If the UE is an eRedCap UE and *intraFreqReselection-eRedCap* in *SIB1* is not available:
      - the UE shall exclude the barred cell as a candidate for cell selection/reselection for 300 seconds.
      - the UE may select another cell on the same frequency if re-selection criteria are fulfilled.
  - If the UE is neither a RedCap UE nor an eRedCap UE; or
  - if the UE is a RedCap UE and *intraFreqReselectionRedCap* in *SIB1* is available; or
  - if the UE is an eRedCap UE and *intraFreqReselection-eRedCap* in *SIB1* is available:

- If the field *intraFreqReselection* in *MIB* message is set to "allowed":
  - the UE may select another cell on the same frequency if re-selection criteria are fulfilled;
  - If the cell is to be treated as if the cell status is "barred" due to no available *SIB1* request configuration for the UE supporting OD-SIB1 as defined in clause 5.2.2.3.1 of TS 38.331 [3]; or
  - If the cell is to be treated as if the cell status is "barred" due to maximum number of PRACH attempts is reached for the UE supporting OD-SIB1 as defined in clause 5.2.2.3.3b of TS 38.331 [3]; or
  - If the cell is to be treated as if the cell status is "barred" due to failing to acquire the *SIB1* upon the expiry of the *SIB1* monitoring window as defined in TS 38.213 [4] for the UE supporting OD-SIB1; or
  - If the cell is to be treated as if the cell status is "barred" due to being unable to acquire the *SIB1*:
    - the UE may exclude the barred cell as a candidate for cell selection/reselection for up to 300 seconds;
  - else:
    - the UE shall exclude the barred cell as a candidate for cell selection/reselection for 300 seconds.
- If the field *intraFreqReselection* in *MIB* message is set to "not allowed":
  - If the cell is to be treated as if the cell status is "barred" due to no available *SIB1* request configuration for the UE supporting OD-SIB1 as defined in clause 5.2.2.3.1 of TS 38.331 [3]; or
  - If the cell is to be treated as if the cell status is "barred" due to maximum number of PRACH attempts is reached for the UE supporting OD-SIB1 as defined in clause 5.2.2.3.3b of TS 38.331 [3]; or
  - If the cell is to be treated as if the cell status is "barred" due to failing to acquire the *SIB1* upon the expiry of the *SIB1* monitoring window as defined in TS 38.213 [4] for the UE supporting OD-SIB1; or
  - If the cell is to be treated as if the cell status is "barred" due to being unable to acquire the *SIB1*:
    - the UE may exclude the barred cell as a candidate for cell selection/reselection for up to 300 seconds;
    - If the cell operates in licensed spectrum:
      - the UE shall not re-select to another cell on the same frequency as the barred cell and exclude such cell(s) as candidate(s) for cell selection/reselection for 300 seconds;
    - else:
      - the UE may select to another cell on the same frequency if the reselection criteria are fulfilled.
  - else:
    - If the cell operates in licensed spectrum, or if this cell belongs to a PLMN which is indicated as being equivalent to the registered PLMN or the selected PLMN of the UE, or if this cell belongs to an SNPN which is equal to or indicated as being equivalent to the registered SNPN or the selected SNPN of the UE:
      - the UE shall not re-select to another cell on the same frequency as the barred cell and exclude such cell(s) as candidate(s) for cell selection/reselection for 300 seconds;
    - else:
      - the UE may select to another cell on the same frequency if the reselection criteria are fulfilled.
    - the UE shall exclude the barred cell as a candidate for cell selection/reselection for 300 seconds.

The cell selection of another cell may also include a change of RAT.

If a UE supporting OD-SIB1 is barred in a cell due to no available *SIB1* request configuration as defined in clause 5.2.2.3.1 of TS 38.331 [3], it considers the cell is no longer barred once *SIB1* request configuration of the cell is acquired.

NOTE 2: If barring of a cell is triggered by the condition of *trackingAreaCode* and *trackingAreaList* not being provided, as specified in TS 38.331 [3], the barring only applies to this PLMN and the UE can re-evaluate the barring condition again due to selection of another PLMN.

5.3.2 Unified access control

The information on cell access restrictions associated with Access Categories and Identities is broadcast in *SIB1* as part of Unified Access Control as specified in TS 38.331 [3].

The UE shall ignore Access Category and Identity related cell access restrictions for cell reselection. A change of the indicated access restriction shall not trigger cell reselection by the UE.

The UE shall consider Access Category and Identity related cell access restrictions for NAS initiated access attempts and RNAU as specified in TS 38.331 [3].

A L2 U2N Relay UE does not need to perform the Unified Access Control as specified in TS 38.331 [3], due to the U2N Remote UE access attempt.

5.4 Tracking Area registration

In the UE, the AS shall report tracking area information to the NAS.

If the UE reads more than one PLMN identity in the current cell, the UE shall report the found PLMN identities that make the cell suitable in the tracking area information to NAS.

If the UE operating in SNPN access mode reads more than one SNPN identity in the current cell, the UE shall report the found SNPN identities that make the cell suitable in the tracking area information to NAS.

The AS of an L2 U2N Remote UE in RRC_IDLE or in RRC_INACTIVE may report the tracking area information to NAS based on the system information received from the PC5-RRC connected L2 Parent UE.

The NAS part of the location registration process is specified in TS 23.122 [9].

5.5 RAN Area registration

The UE performs a RAN-based notification area update (RNAU) periodically or when the UE selects a cell that does not belong to the configured RNA.

A L2 U2N Remote UE, while in RRC_INACTIVE performs RNAU periodically or when the serving cell of the L2 U2N Relay UE changes (e.g., due to reconfiguration with sync, when a different L2 U2N Relay UE is reselected, or when the L2 U2N Relay UE reselects a new cell) and this new serving cell does not belong to the configured RNA of L2 U2N Remote UE.

6 Reception of broadcast information

6.1 Reception of system information

The NAS is informed if the cell selection and reselection results in changes in the received NAS system information.

The UE shall monitor the Paging Occasions (POs) as described in clause 7.1 to receive System Information change notifications in RRC_IDLE and RRC_INACTIVE. The changes in the system information are notified by the network using a Short Message as specified in TS 38.331 [3]. When the Short Message notifies system information changes, then the UE shall acquire or re-acquire the concerned system information as specified in TS 38.331 [3].

A L2 U2N Remote UE when in RRC_IDLE or RRC_INACTIVE may not monitor POs as described in clause 7.1 to receive Short Message when connected with a Parent UE, as specified in TS 38.331 [3].

A L2 U2N Remote UE in RRC_IDLE or RRC_INACTIVE does not receive Short Message from a L2 Parent UE. When receiving a Short Message, the L2 U2N Relay UE may forward to the L2 U2N Remote UE only Public Warning System information (e.g., *SIB6*, *SIB7*, and *SIB8*).

When system information changes, the L2 U2N Remote UE, when in RRC_IDLE or RRC_INACTIVE, relies on the L2 U2N Relay UE or L2 Last U2N Relay UE to acquire or re-acquire the concerned system information and forward them. Further, the L2 U2N Remote UE, when in RRC_CONNECTED, relies on the network to receive concerned system information that has changed.

When system information changes, the L2 Intermediate U2N Relay UE, when in RRC_IDLE or RRC_INACTIVE, may rely on the L2 Parent U2 Relay UE, or on direct reception on the Uu interface if in coverage, to acquire or re-acquire the concerned system information for forwarding to the L2 U2N Remote UE.

6.2 Reception of MBS

A UE receiving or interested to receive MBS broadcast services shall apply the MCCH information acquisition procedure as specified in TS 38.331 [3] to receive the MCCH information. A UE interested to receive MBS broadcast services identifies if a service that it is interested to receive is started or ongoing by receiving the MCCH information, and then receives a MTCH(s) configured using the Broadcast MRB establishment procedure as specified in TS 38.331 [3] and using the DL-SCH reception and MBS broadcast DRX procedure as specified in TS 38.321 [19].

A UE which has joined multicast session(s) and configured to receive MBS multicast services in RRC_INACTIVE state shall apply the multicast MCCH information acquisition procedure as specified in TS 38.331 [3] to receive the multicast MCCH information when UE is in RRC_INACTIVE state and the multicast MCCH is configured in the cell. The UE identifies whether a session is active or not by receiving the indication in *RRCRelease*, multicast MCCH information, or group notification in paging message, and receives the multicast MTCH(s) in RRC_INACTIVE state using the multicast MRB configuration procedure as specified in TS 38.331 [3] and using the DL-SCH reception and MBS multicast DRX procedure as specified in TS 38.321 [19].

UEs which have joined a multicast session(s) and are in RRC_IDLE/RRC_INACTIVE state shall apply the reception of the paging message procedure as specified in TS 38.331 [3] when the UE expects MBS group notification as specified in clause 16.10.5.2 in TS 38.300 [2].

When upper layers provide MBS start time and/or scheduled activation time(s) (as specified in TS23.247 [21]) and the UE has joined an MBS session indicated by TMGI while the UE is in RRC_IDLE or RRC_INACTIVE state, the UE monitors paging as defined in clause 7.1 using the TMGI (as defined in TS 38.331 [3]) during those MBS start time and/or scheduled activation time(s).

NOTE: When the UE is interested to receive MBS broadcast the UE may perform procedures to receive MBS broadcast session(s) as defined in TS 38.331 [3] if upper layer is configured with the MBS start time and/or scheduled activation time(s) (as specified in TS23.247 [21]).

7 Paging

7.1 Discontinuous Reception for paging

The UE may use Discontinuous Reception (DRX) in RRC_IDLE and RRC_INACTIVE state in order to reduce power consumption. The UE monitors one paging occasion (PO) per DRX cycle. A PO is a set of PDCCH monitoring occasions and can consist of multiple time slots (e.g. subframe or OFDM symbol) where paging DCI can be sent (TS 38.213 [4]). One Paging Frame (PF) is one Radio Frame and may contain one or multiple PO(s) or starting point of a PO. A L2 U2N Relay UE monitors the paging occasions of its PC5-RRC connected L2 U2N Remote UEs. In this case, the DRX cycle and UE ID mentioned in this clause refer to those of the L2 U2N Remote UE. A L2 U2N Last Relay UE monitors the paging occasions of its downstream L2 U2N Remote UEs. In this case, the DRX cycle and UE ID mentioned in this clause refer to those of the L2 U2N Remote UE.

In multi-beam operations, the UE assumes that the same paging message and the same Short Message are repeated in all transmitted beams and thus the selection of the beam(s) for the reception of the paging message and Short Message is up to UE implementation. The paging message is same for both RAN initiated paging and CN initiated paging.

The UE initiates RRC Connection Resume procedure upon receiving RAN initiated paging. If the UE receives a CN initiated paging in RRC_INACTIVE state, the UE moves to RRC_IDLE and informs NAS. However, if a L2 U2N Relay UE in RRC_INACTIVE state receives a CN initiated paging for a L2 U2N Remote UE, the L2 U2N Relay UE does not move to RRC_IDLE state.

NOTE 0a: The L2 U2N Remote UE does not need to monitor the PO in order to receive the paging message.

NOTE 0b: While the SDT procedure is ongoing in RRC_INACTIVE state, the UE monitors the PO in order to receive only the Short Message as specified in TS 38.331 [3].

The PF and PO for paging are determined by the following formulae:

SFN for the PF is determined by:

(SFN + PF_offset) mod T = (T div N)*(UE_ID mod N)

Index (i_s), indicating the index of the PO is determined by:

i_s = floor (UE_ID/N) mod Ns

The PDCCH monitoring occasions for paging are determined according to *pagingSearchSpace* as specified in TS 38.213 [4] and *firstPDCCH-MonitoringOccasionOfPO* (or *pagingAdaptFirstPDCCH-MonitoringOccasionOfPO* for paging adaptation) and *nrofPDCCH-MonitoringOccasionPerSSB-InPO* if configured as specified in TS 38.331 [3]. When *SearchSpaceId* = 0 is configured for *pagingSearchSpace*, the PDCCH monitoring occasions for paging are same as for RMSI as defined in clause 13 in TS 38.213 [4].

When *SearchSpaceId* = 0 is configured for *pagingSearchSpace*, Ns is either 1 or 2. For Ns = 1, there is only one PO which starts from the first PDCCH monitoring occasion for paging in the PF. For Ns = 2, PO is either in the first half frame (i_s = 0) or the second half frame (i_s = 1) of the PF.

When *SearchSpaceId* other than 0 is configured for *pagingSearchSpace*, the UE monitors the $(i_s + 1)^{th}$ PO. A PO is a set of 'S*X ' consecutive PDCCH monitoring occasions where 'S' is the number of actual transmitted SSBs determined according to *ssb-PositionsInBurst* in *SIB1* and X is the *nrofPDCCH-MonitoringOccasionPerSSB-InPO* if configured or is equal to 1 otherwise. The $[x*S+K]^{th}$ PDCCH monitoring occasion for paging in the PO corresponds to the Kth transmitted SSB, where x=0,1,…,X-1, K=1,2,…,S. The PDCCH monitoring occasions for paging which do not overlap with UL symbols (determined according to *tdd-UL-DL-ConfigurationCommon*) are sequentially numbered from zero starting from the first PDCCH monitoring occasion for paging in the PF. When *firstPDCCH-MonitoringOccasionOfPO* (or *pagingAdaptFirstPDCCH-MonitoringOccasionOfPO* for paging adaptation) is present, the starting PDCCH monitoring occasion number of $(i_s + 1)^{th}$ PO is the $(i_s + 1)^{th}$ value of the *firstPDCCH-MonitoringOccasionOfPO* (or *pagingAdaptFirstPDCCH-MonitoringOccasionOfPO* for paging adaptation) parameter; otherwise, it is equal to i_s * S*X. If X > 1, when the UE detects a PDCCH transmission addressed to P-RNTI within its PO, the UE is not required to monitor the subsequent PDCCH monitoring occasions for this PO.

NOTE 1: A PO associated with a PF may start in the PF or after the PF.

NOTE 2: The PDCCH monitoring occasions for a PO can span multiple radio frames. When *SearchSpaceId* other than 0 is configured for *paging-SearchSpace* the PDCCH monitoring occasions for a PO can span multiple periods of the paging search space.

The following parameters are used for the calculation of PF and i_s above:

T: DRX cycle of the UE.

If the UE does not operate in eDRX as defined in clause 7.4:

- T is determined by the shortest of the UE specific DRX value configured by RRC (if any), the UE specific DRX value configured by upper layers (if any), and a default DRX value broadcast in system information. For L2 U2N Relay UE, T for a L2 U2N Remote UE is determined by the shortest of the UE specific DRX value provided in PC5-RRC signalling and a default DRX value broadcast in system information.

In RRC_IDLE state, if the UE operates in eDRX and eDRX is configured by upper layers, i.e., $T_{eDRX, CN}$, according to clause 7.4:

- If $T_{eDRX, CN}$ is no longer than 1024 radio frames:
  - $T = T_{eDRX, CN}$;
- else:
  - During CN configured PTW, T is determined by the shortest of UE specific DRX value, if configured by upper layers, and the default DRX value broadcast in system information.

In RRC_INACTIVE state, if the UE operates in eDRX and eDRX is configured by RRC, i.e., $T_{eDRX, RAN}$ (if any), and upper layers, i.e., $T_{eDRX, CN}$, as defined in clause 7.4:

- If both $T_{eDRX, CN}$ and used $T_{eDRX, RAN}$ are no longer than 1024 radio frames:
  - $T = \min\{T_{eDRX, RAN}, T_{eDRX, CN}\}$.
- If $T_{eDRX, CN}$ is no longer than 1024 radio frames and no $T_{eDRX, RAN}$ is configured or used:
  - T is determined by the shortest of UE specific DRX value configured by RRC and $T_{eDRX, CN}$.
- If $T_{eDRX, CN}$ is longer than 1024 radio frames:
  - If $T_{eDRX, RAN}$ is not configured or used:
    - During CN configured PTW, T is determined by the shortest of the UE specific DRX value configured by RRC, the UE specific DRX value configured by upper layers (if any), and a default DRX value broadcast in system information. Outside the CN configured PTW, T is determined by the UE specific DRX value configured by RRC;
  - else if used $T_{eDRX, RAN}$ is no longer than 1024 radio frames:
    - During CN configured PTW, T is determined by the shortest of the UE specific DRX value, if configured by upper layers and $T_{eDRX, RAN}$, and a default DRX value broadcast in system information. Outside the CN configured PTW, T is determined by $T_{eDRX, RAN;}$
  - else if used $T_{eDRX, RAN}$ is longer than 1024 radio frames:
    - During the overlapped part of CN configured PTW and RAN configured PTW, T is determined by the shortest of the UE specific DRX value configured by RRC, the UE specific DRX value configured by upper layers (if any), and a default DRX value broadcast in system information;
    - During CN configured PTW and outside RAN configured PTW, T is determined by the shortest of the UE specific DRX value configured by upper layers (if any), and a default DRX value broadcast in system information;
    - Outside CN configured PTW and during RAN configured PTW, T is determined by the UE specific DRX value configured by RRC.

N: number of total paging frames in T

Ns: number of paging occasions for a PF

PF_offset: offset used for PF determination

UE_ID:

If the UE operates in eDRX as specified in clause 7.4:

- 5G-S-TMSI mod 4096

else:

- 5G-S-TMSI mod 1024

Parameters *Ns*, *nAndPagingFrameOffset*, *nrofPDCCH-MonitoringOccasionPerSSB-InPO*, and the length of default DRX Cycle are signaled in *SIB1*. The values of N and PF_offset are derived from the parameter *nAndPagingFrameOffset* as defined in TS 38.331 [3]. The parameter *firstPDCCH-MonitoringOccasionOfPO* is signalled in *SIB1* for paging in the BWP configured by *initialDownlinkBWP*. For paging in a DL BWP other than the BWP configured by *initialDownlinkBWP*, the parameter *first-PDCCH-MonitoringOccasionOfPO* is signaled in the corresponding BWP configuration.

For a UE in RRC_IDLE or RRC_INACTIVE state and supporting paging adaptation, if *pagingAdapt-NS* and *pagingAdaptNAndPagingFrameOffset* are signaled in *SIB1*, the UE determines the value of Ns from *pagingAdapt-NS*, N and PF_offset from the parameter *pagingAdaptNAndPagingFrameOffset* as defined in TS 38.331 [3], and only monitors the PO(s) derived from these paging parameters. The parameter *pagingAdaptFirstPDCCH-MonitoringOccasionOfPO* for paging adaptation is signalled in *SIB1* for paging in the BWP configured by *initialDownlinkBWP*. For paging in a DL BWP other than the BWP configured by *initialDownlinkBWP*, the parameter *pagingAdaptFirstPDCCH-MonitoringOccasionOfPO* for paging adaptation is signaled in the corresponding BWP configuration.

If the UE has no 5G-S-TMSI, for instance when the UE has not yet registered onto the network, the UE shall use as default identity UE_ID = 0 in the PF and i_s formulas above.

5G-S-TMSI is a 48 bit long bit string as defined in TS 23.501 [10]. 5G-S-TMSI shall in the formulae above be interpreted as a binary number where the left most bit represents the most significant bit.

In RRC_INACTIVE state, if the UE supports *inactiveStatePO-Determination* and the network broadcasts *ranPagingInIdlePO* with value "true", the UE shall use the same i_s as for RRC_IDLE state. Otherwise, the UE determines the i_s based on the parameters and formula above.

In RRC_INACTIVE state, if used eDRX value configured by upper layers is no longer than 1024 radio frames, the UE shall use the same i_s as for RRC_IDLE state.

In RRC_INACTIVE state, if used eDRX value configured by upper layers is longer than 1024 radio frames, during CN PTW, the UE shall use the same i_s as for RRC_IDLE state. Outside CN PTW, the UE shall use the i_s for RRC_INACTIVE state.

7.2 Paging Early Indication

7.2.1 Paging Early Indication reception

The UE may use Paging Early Indication (PEI) in RRC_IDLE and RRC_INACTIVE states in order to reduce power consumption. If PEI configuration is provided in system information, the UE in RRC_IDLE or RRC_INACTIVE state supporting PEI (except for the UEs expecting MBS group notification) can monitor PEI using PEI parameters in system information according to the procedure described below.

If *lastUsedCellOnly* is configured in system information of a cell, the UE monitors PEI in this cell only if the UE most recently received *RRCRelease* without *noLastCellUpdate* in this cell. Otherwise (i.e., if *lastUsedCellOnly* is not configured in system information of a cell), the UE monitors PEI in the camped cell.

The UE monitors one PEI occasion per DRX cycle. A PEI occasion (PEI-O) is a set of PDCCH monitoring occasions (MOs) and can consist of multiple time slots (e.g. subframes or OFDM symbols) where PEI can be sent (TS 38.213 [4]). In multi-beam operations, the UE assumes that the same PEI is repeated in all transmitted beams and thus the selection of the beam(s) for the reception of the PEI is up to UE implementation.

The time location of PEI-O for UE's PO is determined by a reference point and an offset:

- The reference point is the start of a reference frame determined by a frame-level offset from the start of the first PF of the PF(s) associated with the PEI-O, provided by *pei-FrameOffset* in SIB1;
- The offset is a symbol-level offset from the reference point to the start of the first PDCCH MO of this PEI-O, provided by *firstPDCCH-MonitoringOccasionOfPEI-O* in SIB1.

If one PEI-O is associated with POs of two PFs, the two PFs are consecutive PFs calculated by the parameters *PF_offset*, *T*, *Ns*, and *N*. The first PF of the PFs associated with the PEI-O is provided by (SFN for PF) - floor

(i_{PO}/*Ns*)**T*/*N*, where SFN for PF is determined in clause 7.1, i_{PO} is defined in clause 10.4a in TS 38.213[4], *T*, *Ns*, and *N* are determined in clause 7.1.

The PDCCH MOs for PEI are determined as specified in TS 38.213 [4] according to *pei-SearchSpace*, *pei-FrameOffset*, *firstPDCCH-MonitoringOccasionOfPEI-O* and *nrofPDCCH-MonitoringOccasionPerSSB-InPO* if configured as specified in TS 38.331 [3]. When *SearchSpaceId* = 0 is configured for *pei-SearchSpace*, the PDCCH MOs for PEI are same as for RMSI as defined in clause 13 in TS 38.213 [4]. UE determines first PDCCH MO for PEI-O based on *pei-FrameOffset* and *firstPDCCH-MonitoringOccasionOfPEI-O*, as for the case with *SearchSpaceId* > 0 configured.

When *SearchSpaceId* = 0 is configured for *pei-SearchSpace*, the UE monitors the PEI-O according to *searchSpaceZero*. When *SearchSpaceId* other than 0 is configured for *pei-SearchSpace*, the UE monitors the PEI-O according to the search space with the configured *SearchSpaceId*.

A PEI occasion is a set of 'S*X' consecutive PDCCH MOs, where 'S' is the number of actual transmitted SSBs determined according to *ssb-PositionsInBurst* in *SIB1*, and X is the *nrofPDCCH-MonitoringOccasionPerSSB-InPO* if configured or is equal to 1 otherwise. The $[x*S+K]^{th}$ PDCCH MO for PEI in the PEI-O corresponds to the Kth transmitted SSB, where x=0,1,…,X-1, K=1,2,…,S. The PDCCH MOs for PEI which do not overlap with UL symbols (determined according to *tdd-UL-DL-ConfigurationCommon*) are sequentially numbered from zero starting from the first PDCCH MO for PEI in the PEI-O. When the UE detects a PEI within its PEI-O, the UE is not required to monitor the subsequent MO(s) associated with the same PEI-O.

If the UE detects PEI and the PEI indicates the subgroup the UE belongs to monitor its associated PO, as specified in clause 10.4a in TS 38.213 [4], the UE monitors the associated PO as specified in clause 7.1. If the UE does not detect PEI on the monitored PEI occasion or the PEI does not indicate the subgroup the UE belongs to monitor its associated PO, as specified in clause 10.4a in TS 38.213 [4], the UE is not required to monitor the associated PO as specified in clause 7.1.

If the UE is unable to monitor the PEI occasion (i.e. all valid PDCCH MO for PEI) corresponding to its PO, e.g. during cell re-selection, the UE monitors the associated PO according to clause 7.1.

In RRC_INACTIVE state, when the UE uses the same i_s as for RRC_IDLE state as specified in clause 7.1, the UE shall use the same i_{PO} as for RRC_IDLE state. Otherwise, the UE determines the i_{PO} based on the formula defined in clause 10.4a in TS 38.213 [4].

For a UE supporting paging adaptation and PEI, if *pagingAdaptPEI-Config* is signaled in system information, the UE in RRC_IDLE or RRC_INACTIVE state can monitor the PEI occasion according to *pagingAdaptPEI-Config*, *pei-FrameOffset-r19* and *pagingAdaptFirstPDCCH-MonitoringOccasionOfPEI-O*.

7.3 Subgrouping for PEI

7.3.0 General

If PEI and subgrouping are configured, UEs monitoring the same PO can be divided into one or more subgroups. With subgrouping, the UE monitors the associated PO if the corresponding bit for subgroup the UE belongs to is indicated as 1 by PEI corresponding to its PO, as specified in clause 10.4a in TS 38.213 [4].

The following parameters are used for the determination of subgroup ID:

- *subgroupsNumPerPO*: total number of subgroups for both CN assigned subgrouping (if any) and UE_ID based subgrouping (if any) in a PO, which is broadcasted in system information;
- *subgroupsNumForUEID*: number of subgroups for UE_ID based subgrouping in a PO, which is broadcasted in system information.

UE's subgroup can be either assigned by CN as specified in clause 7.3.1 or formed based on UE_ID as specified in clause 7.3.2:

- If *subgroupsNumForUEID* is absent in *subgroupConfig*, the subgroup ID based on CN assigned subgrouping as specified in clause 7.3.1, if available for the UE, is used in the cell.

- If both *subgroupsNumPerPO* and *subgroupsNumForUEID* are configured, and *subgroupsNumForUEID* has the same value as *subgroupsNumPerPO*, the subgroup ID based on UE_ID based subgrouping as specified in clause 7.3.2 is used in the cell.
- If both *subgroupsNumPerPO* and *subgroupsNumForUEID* are configured, and *subgroupsNumForUEID* < *subgroupsNumPerPO*:
  - The subgroup ID based on CN assigned subgrouping as specified in clause 7.3.1, if available for the UE, is used in the cell;
- Otherwise, the subgroup ID based on UE_ID based subgrouping as specified in clause 7.3.2 is used in the cell.

If a UE has no CN assigned subgroup ID or does not support CN assigned subgrouping, and there is no configuration for *subgroupsNumForUEID*, the UE monitors the associated PO according to clause 7.1.

7.3.1 CN assigned subgrouping

Paging with CN assigned subgrouping is used in the cell which supports CN assigned subgrouping, as described in clause 7.3.0. A UE supporting CN assigned subgrouping in RRC_IDLE or RRC_INACTIVE state can be assigned a subgroup ID (between 0 to 7) by AMF through NAS signalling. The UE belonging to the assigned subgroup ID monitors its associated PEI which indicates the paged subgroup(s) as specified in clause 7.2.

7.3.2 UE_ID based subgrouping

Paging with UE_ID based subgrouping is used in the cell which supports UE_ID based subgrouping, as described in clause 7.3.0.

If the UE is not configured with a CN assigned subgroup ID, or if the UE configured with a CN assigned subgroup ID is in a cell supporting only UE_ID based subgrouping, the subgroup ID of the UE is determined by the formula below:

SubgroupID = (floor(UE_ID/(N*Ns)) mod subgroupsNumForUEID) + (subgroupsNumPerPO - subgroupsNumForUEID),

where:

N: number of total paging frames in T, which is the DRX cycle of RRC_IDLE state as specified in clause 7.1

Ns: number of paging occasions for a PF

UE_ID: 5G-S-TMSI mod X, where X is 32768, if eDRX is applied; otherwise, X is 8192

subgroupsNumForUEID: number of subgroups for UE_ID based subgrouping in a PO, which is broadcasted in system information

In RRC_INACTIVE state with CN configured PTW the SubgroupID used outside CN PTW is the same as the SubgroupID used inside CN PTW.

The UE belonging to the SubgroupID monitors its associated PEI which indicates the paged subgroup(s) as specified in clause 7.2.

7.4 Paging in extended DRX

The UE may be configured by upper layers and/or RRC with an extended DRX (eDRX) cycle $T_{eDRX, CN}$ and/or $T_{eDRX, RAN}$.

For CN paging, the UE operates in eDRX in RRC_IDLE or RRC_INACTIVE states if the UE is configured for eDRX by upper layers and *eDRX-AllowedIdle* is signalled in SIB1; otherwise, the UE does not operate in eDRX.

For RAN paging, the UE in RRC_INACTIVE state:

- if the UE is configured for eDRX by *ran-ExtendedPagingCycleConfig-r18* and *eDRX-AllowedInactive-r18* is signalled in SIB1:

  - operates in eDRX with an eDRX cycle $T_{eDRX, RAN}$ configured by *extendedPagingCycle-r18*;

- else if the UE is configured for eDRX by *ran-ExtendedPagingCycle-r17* and *eDRX-AllowedInactive-r17* is signalled in SIB1:

  - operates in eDRX with an eDRX cycle $T_{eDRX, RAN}$ configured by *ran-ExtendedPagingCycle-r17*;

- else:

  - does not operate in eDRX.

If the UE operates in eDRX with an eDRX cycle no longer than 1024 radio frames, it monitors POs as defined in 7.1 with configured eDRX cycle. Otherwise, a UE operating in eDRX monitors POs as defined in 7.1 during a periodic Paging Time Window (PTW) configured for the UE. The PTW is UE-specific and is determined by a Paging Hyperframe (PH), a starting position within the PH (PTW_start) and an ending position (PTW_end). PH, PTW_start and PTW_end are given by the following formula:

The PH for CN is the H-SFN satisfying the following equations:

H-SFN mod $T_{eDRX, CN}$= (UE_ID_H mod $T_{eDRX, CN}$), where

- $T_{eDRX, CN}$: UE-specific eDRX cycle in Hyper-frames, ($T_{eDRX, CN}$ = 2, …, 1024 Hyper-frames) configured by upper layers.

The PH for RAN is the H-SFN satisfying the following equations:

H-SFN mod T_{eDRX_RAN}= (UE_ID_H mod T_{eDRX_RAN}), where

- T_{eDRX_RAN}: UE-specific eDRX cycle in Hyper-frames, (T_{eDRX_RAN} = 2, …, 1024 Hyper-frames) configured by RRC.

For CN configured PTW:

PTW_start denotes the first radio frame of the PH for CN that is part of the PTW and has SFN satisfying the following equation:

SFN = 128 * $i_{eDRX, CN}$, where

- $i_{eDRX, CN}$ = floor(UE_ID_H /$T_{eDRX, CN}$) mod 8

PTW_end is the last radio frame of the PTW and has SFN satisfying the following equation:

SFN = (PTW_start + L*100 - 1) mod 1024, where

- L = Paging Time Window (PTW) length (in seconds) configured by upper layers

For RAN configured PTW:

PTW_start denotes the first radio frame of the PH for RAN that is part of the PTW and has SFN satisfying the following equation:

SFN = 128 * i_{eDRX_CN}, where

- i_{eDRX_CN} = floor(UE_ID_H /T_{eDRX_CN}) mod 8

PTW_end is the last radio frame of the PTW and has SFN satisfying the following equation:

SFN = (PTW_start + L*100 - 1) mod 1024, where

- L = Paging Time Window (PTW) length (in seconds) configured by RRC

UE_ID_H is defined as follows:

UE_ID_H: 13 most significant bits of the Hashed ID.

Hashed ID is defined as follows:

Hashed_ID is Frame Check Sequence (FCS) for the bits b31, b30…, b0 of 5G-S-TMSI.

5G-S-TMSI = <b47, b46, …, b0> as defined in TS 23.003 [23].

The 32-bit FCS shall be the ones complement of the sum (modulo 2) of Y1 and Y2, where

- Y1 is the remainder of x^k ($x^{31} + x^{30} + x^{29} + x^{28} + x^{27} + x^{26} + x^{25} + x^{24} + x^{23} + x^{22} + x^{21} + x^{20} + x^{19} + x^{18} + x^{17} + x^{16} + x^{15} + x^{14} + x^{13} + x^{12} + x^{11} + x^{10} + x^9 + x^8 + x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + x^1 + 1$) divided (modulo 2) by the generator polynomial $x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$, where k is 32; and

- Y2 is the remainder of Y3 divided (modulo 2) by the generator polynomial $x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$, where Y3 is the product of x^{32} by "b31, b30…, b0 of S-TMSI or 5G-S-TMSI", i.e., Y3 is the generator polynomial x^{32} ($b31*x^{31} + b30*x^{30} + \ldots + b0*1$).

NOTE: The Y1 is 0xC704DD7B for any 5G-S-TMSI value. An example of hashed ID calculation is in Annex A.

7.5 LP-WUS monitoring

7.5.0 General

The UE supporting LP-WUS may monitor LP-WUS in RRC_IDLE and RRC_INACTIVE states in order to reduce power consumption. If LP-WUS configuration is provided in system information, the UE in RRC_IDLE or RRC_INACTIVE state supporting LP-WUS (except for the UEs expecting MBS group notification) may start LP-WUS monitoring using LP-WUS parameters in system information according to the procedure described below if the entry condition in clause 7.5.1 is fulfilled. The UE may stop LP-WUS monitoring and follows the paging monitoring procedure as described in clause 7.1 or 7.2 if the exit condition in clause 7.5.1 is fulfilled. LP-WUS monitoring can be disabled in the UE via NAS signalling as specified in TS 23.501 [10]. If LP-WUS disabled indication via NAS signalling is absent, LP-WUS monitoring is enabled.

If the UE detects LP-WUS and the LP-WUS is associated with the UE as specified in clause 10.4C in TS 38.213 [4], the UE monitors the associated PO as specified in clause 7.1 or monitors PEI as specified in clause 7.2 if PEI is supported and PEI configuration is provided in system information, which is up to UE implementation. If the UE does not detect an LP-WUS on the monitored LP-WUS occasion (LO) or the LP-WUS is not associated with the UE as specified in clause 10.4C in TS 38.213 [4], the UE is not required to monitor the associated PO as specified in clause 7.1.

The UE monitors one LO per DRX cycle. An LO is a set of LP-WUS monitoring occasions (LP-WUS MOs). In multi-beam operations, the UE assumes that the same LP-WUS is repeated in all transmitted beams for which LP-WUS is configured and thus the selection of the beam(s) for the reception of the LP-WUS is up to UE implementation.

The time location of an LO for UE's PO is determined by a reference PO/PF and the configured frame-level offset:

- The reference point (reference PO/PF) is the start of the PF, or the first PF of the PF(s) (if mapping of POs from multiple PFs to one LO is configured), associated with the LO. The reference PF for the LO of a PO is provided by (SFN for PF) – floor(i_{PO}/N_S) *T/N, where SFN for PF is determined in clause 7.1, i_{PO} is defined in clause 10.4C in TS 38.213 [4], T, Ns, and N are determined in clause 7.1.

- The frame-level offset between the LO and the reference PO/PF is provided by *lpwus-LO-FrameOffsetList* in SIB1. If *lpwus-PO-NumPerLO* is less than *Ns*, the UE uses the (floor(*i_s*/*lpwus-PO-NumPerLO*)+1)-th value in *offsetForLongerWakeUpDelay* (if configured) or the (floor(*i_s*/*lpwus-PO-NumPerLO*)+1)-th value in *offsetForShorterWakeUpDelay* (if configured) provided by *lpwus-LO-FrameOffsetList*.

When only one of *offsetForLongerWakeUpDelay* and *offsetForShorterWakeUpDelay* is configured in *lpwus-LO-FrameOffsetList*:

- If the gap between the end of the last LP-WUS MO the UE would monitor in the LO associated with the value in *offsetForLongerWakeUpDelay* or *offsetForShorterWakeUpDelay* (whichever is configured) and the start of the corresponding PO is no less than the wake-up delay that a UE supports, the UE monitors the LO associated with the offset;

- Otherwise, the UE follows the paging monitoring procedure as described in clause 7.1 or 7.2.

When both *offsetForLongerWakeUpDelay* and *offsetForShorterWakeUpDelay* are configured in *lpwus-LO-FrameOffsetList*:

- If the gap between the end of the last LP-WUS MO the UE would monitor in the LO associated with the value in *offsetForShorterWakeUpDelay* and the start of the corresponding PO is no less than the wake-up delay a UE reports, the UE monitors the LO associated with the value in *offsetForShorterWakeUpDelay*;

- Otherwise if the gap between the end of the last LP-WUS MO the UE would monitor in the LO associated with the value in *offsetForLongerWakeUpDelay* and the start of the corresponding PO is no less than the wake-up delay a UE reports, the UE monitors the LO associated with the value in *offsetForLongerWakeUpDelay*;

- Otherwise, the UE follows the paging monitoring procedure as described in clause 7.1 or 7.2.

If the UE in RRC_IDLE and RRC_INACTIVE is unable to monitor the LP-WUS occasion in all MO corresponding to its PO, the UE follows the paging monitoring procedure as described in clause 7.1 or 7.2.

In RRC_INACTIVE state, when the UE uses the same i_s as for RRC_IDLE state as specified in clause 7.1, the UE shall use the same i_{PO} for LP-WUS as for RRC_IDLE state. Otherwise, the UE determines the i_{PO} for LP-WUS based on the i_s for RRC_INACTIVE state.

For a UE supporting paging adaptation and LP-WUS, if *lpwus-LO-FrameOffsetListForPagingAdapt* is signaled in system information, the UE in RRC_IDLE or RRC_INACTIVE state can monitor the LP-WUS occasion according to *lpwus-LO-FrameOffsetListForPagingAdapt*.

7.5.1 Condition for LP-WUS monitoring

The entry condition for LP-WUS monitoring is fulfilled when:

- Srxlev > $S_{LP_WUS_EntryThresholdP_MR}$, and,

- $Q_{rxlevmeas_lr}$ > $Q_{LP_WUS_EntryThresholdP_LR}$, if $Q_{LP_WUS_EntryThresholdP_LR}$ is configured, and,

- Squal > $S_{LP_WUS_EntryThresholdQ_MR}$, if $S_{LP_WUS_EntryThresholdQ_MR}$ is configured, and

- $Q_{qualmeas_lr}$ > $Q_{LP_WUS_EntryThresholdQ_LR}$, if $Q_{LP_WUS_EntryThresholdQ_LR}$ is configured.

The exit condition for LP-WUS monitoring is fulfilled when:

- $Q_{rxlevmeas_lr}$ < $Q_{LP_WUS_ExitThresholdP_LR}$ or,

- $Q_{qualmeas_lr}$ < $Q_{LP_WUS_ExitThresholdQ_LR}$, if $Q_{LP_WUS_ExitThresholdQ_LR}$ is configured.

Where:

- Srxlev = current Srxlev value of the serving cell (dB).

- Squal = current Squal value of the serving cell (dB).

- $Q_{rxlevmeas_lr}$= current measured cell RX level value of the serving cell based on LR (RSRP).

- $Q_{qualmeas_lr}$ = current measured cell quality value of the serving cell based on LR (RSRQ).

- $S_{LP_WUS_EntryThresholdP_MR}$: The Srxlev threshold for entry condition of LP-WUS monitoring based on MR.

- $S_{LP_WUS_EntryThresholdQ_MR}$: The Squal threshold for entry condition of LP-WUS monitoring based on MR.

- $Q_{LP_WUS_EntryThresholdP_LR}$: The cell RX level threshold for entry condition of LP-WUS monitoring based on LR.

- $Q_{LP_WUS_EntryThresholdQ_LR}$: The cell quality threshold for entry condition of LP-WUS monitoring based on LR.

- $Q_{LP_WUS_ExitThresholdP_LR}$: The cell RX level threshold for exit condition of LP-WUS monitoring based on LR.

- $Q_{LP_WUS_ExitThresholdQ_LR}$: The cell quality threshold for exit condition of LP-WUS monitoring based on LR.

These thresholds can be configured separately for LR measurements based on LP-SS and LR measurements based on SSB if a cell supports both measurement types as specified in TS 38.331 [3]. $S_{LP_WUS_EntryThresholdP_MR}$ is the parameter $S_{ThresholdP1}$ or $S_{ThresholdP2}$ in TS 38.331 [3]. $S_{LP_WUS_EntryThresholdQ_MR}$ is the parameter $S_{ThresholdQ1}$ or $S_{ThresholdQ2}$ in TS 38.331 [3]. $Q_{LP_WUS_EntryThresholdP_LR}$ is the parameter $S_{ThresholdP1\text{-}LR}$ or $S_{ThresholdP3\text{-}LR}$ in TS 38.331 [3]. $Q_{LP_WUS_EntryThresholdQ_LR}$ is the parameter $S_{ThresholdQ1\text{-}LR}$ or $S_{ThresholdQ3\text{-}LR}$ in TS 38.331 [3]. $Q_{LP_WUS_ExitThresholdP_LR}$ is the parameter $S_{ThresholdP2\text{-}LR}$ or $S_{ThresholdP4\text{-}LR}$ in TS 38.331 [3]. $Q_{LP_WUS_ExitThresholdQ_LR}$ is the parameter $S_{ThresholdQ2\text{-}LR}$ or $S_{ThresholdQ4\text{-}LR}$ in TS 38.331 [3]. If UE supports both measurement types, it is up to UE implementation to choose LR measurements based on LP-SS or based on SSB for the determination of the LP-WUS monitoring entry/exit conditions.

In multi-beam operations, measured cell RX level value and measured cell quality value of the serving cell based on LR is up to UE implementation.

7.6 Subgrouping for LP-WUS

7.6.0 General

With LP-WUS monitoring, UEs monitoring the same PO can be divided into one or more subgroups.

The following parameters are used for the determination of subgroup ID for LP-WUS:

- *lp-SubgroupsNumPerPO*: total number of subgroups for both CN assigned subgrouping (if any) and UE_ID based subgrouping (if any) for LP-WUS in a PO, which is broadcasted in system information;

- *lp-SubgroupsNumForUEID*: number of subgroups for UE_ID based subgrouping for LP-WUS in a PO, which is broadcasted in system information.

UE's subgroup for LP-WUS can be either assigned by CN as specified in clause 7.6.1 or formed based on UE_ID as specified in clause 7.6.2:

- If *lp-SubgroupsNumForUEID* is absent in *lp-SubgroupConfig*, the subgroup ID based on CN assigned subgrouping for LP-WUS as specified in clause 7.6.1, if available for the UE, is used in the cell.

- If both *lp-SubgroupsNumPerPO* and *lp-SubgroupsNumForUEID* are configured, and *lp-SubgroupsNumForUEID* has the same value as *lp-SubgroupsNumPerPO*, the subgroup ID based on UE_ID based subgrouping for LP-WUS as specified in clause 7.6.2 is used in the cell.

- If both *lp-SubgroupsNumPerPO* and *lp-SubgroupsNumForUEID* are configured, and *lp-SubgroupsNumForUEID* < *lp-SubgroupsNumPerPO*:

  - The subgroup ID based on CN assigned subgrouping for LP-WUS as specified in clause 7.6.1, if available for the UE, is used in the cell;

- Otherwise, the subgroup ID based on UE_ID based subgrouping for LP-WUS as specified in clause 7.6.2 is used in the cell.

If a UE has no CN assigned subgroup ID for LP-WUS or does not support CN assigned subgrouping for LP-WUS, and there is no configuration for *lp-SubgroupsNumForUEID*, the UE monitors the associated PO according to clause 7.1 or monitors PEI as specified in clause 7.2.

7.6.1 CN assigned subgrouping

LP-WUS with CN assigned subgrouping is used in the cell which supports CN assigned subgrouping for LP-WUS, as described in clause 7.6.0. A UE supporting CN assigned subgrouping for LP-WUS in RRC_IDLE or RRC_INACTIVE state can be assigned a subgroup ID (between 0 to 30) by AMF through NAS signalling. The UE belonging to the assigned subgroup ID monitors its associated LP-WUS as specified in clause 7.5.

7.6.2 UE_ID based subgrouping

LP-WUS with UE_ID based subgrouping is used in the cell which supports UE_ID based subgrouping for LP-WUS, as described in clause 7.6.0.

If the UE is not configured with a CN assigned subgroup ID for LP-WUS, or if the UE configured with a CN assigned subgroup ID for LP-WUS is in a cell supporting only UE_ID based subgrouping for LP-WUS, the subgroup ID of the UE for LP-WUS is determined by the formula below:

lp-SubgroupID = (floor(UE_ID/(N*Ns*Np)) mod lp-SubgroupsNumForUEID) + (lp-SubgroupsNumPerPO – lp-SubgroupsNumForUEID),

where:

N: number of total paging frames in T, which is the DRX cycle of RRC_IDLE state as specified in clause 7.1

Ns: number of paging occasions for a PF

Np is the number of *subgroupsNumForUEID* for PEI, if broadcasted in system information and UE supports PEI; otherwise, Np is 1

UE_ID: 5G-S-TMSI mod 1048576

lp-SubgroupsNumForUEID and *lp-SubgroupsNumPerPO* are the subgroup number for UE_ID based subgrouping for LP-WUS and the total subgroup number for LP-WUS, respectively.

In RRC_INACTIVE state with CN configured PTW, the SubgroupID for LP-WUS used outside CN PTW is the same as the SubgroupID used inside CN PTW.

The UE belonging to the subgroup ID monitors its associated LP-WUS as specified in clause 7.5.

8 Sidelink Operation

8.1 NR sidelink communication, and V2X sidelink communication, NR sidelink discovery, and ranging/ sidelink positioning

The UE may transmit or receive NR sidelink communication/discovery if it fulfils the condition(s) defined in TS 38.331 [3], clause 5.8.2. When UE is in-coverage for sidelink operation as defined in clause 8.2, the UE may perform NR sidelink communication/discovery according to *SIB12*, and when out-of-coverage for sidelink, the UE may perform NR sidelink communication/discovery according to *SL-PreconfigurationNR* or according to *SIB12* of the cell on the frequency which provides inter-carrier NR sidelink configuration, or according to *SIB12* received from the connected L2 U2N Relay UE as specified in TS 38.331 [3]. The UE shall not perform NR sidelink communication/discovery according to *SL-PreconfigurationNR* if the UE detects a cell providing NR sidelink configuration or inter-carrier NR sidelink configuration for the frequency UE is interested to perform NR sidelink communication/discovery on, or if the UE is a L2 U2N Remote UE and has received *SIB12* from the connected L2 U2N Relay UE.

The UE may transmit or receive V2X sidelink communication if it fulfills the condition(s) defined in TS 36.331[6], clause 5.10.1d. When UE is in-coverage for sidelink operation as defined in clause 8.2, the UE may perform V2X sidelink communication according to *SIB13/ SIB14* of the cell on an NR frequency.

The UE may transmit or receive SL-PRS for ranging/sidelink positioning if it fulfils the conditions defined in TS 38.331 [3].

The U2N Remote UE, the U2N Relay UE, the U2U Remote UE, or the U2U Relay UE may transmit NR sidelink relay discovery (i.e., as specified in TS 23.304 [22]) if it fulfills the condition(s) defined in TS 38.331 [3].

For NR sidelink broadcast and groupcast, the UE may obtain SL DRX configuration from *SIB12* (for in-coverage UE, as defined in clause 8.2, in RRC_IDLE and RRC_INACTIVE state; or for out-of-coverage UE, as defined in clause 8.2,

on the frequency which the UE is configured to perform NR sidelink communication/discovery and which is included in *sl-FreqInfoList*/*sl-FreqInfoListSizeExt* in *SIB12*) or *SL-PreconfigurationNR* (for out-of-coverage UE, as defined in clause 8.2, on the frequency which the UE is configured to perform NR sidelink communication/discovery and which is not included in *sl-FreqInfoList*/*sl-FreqInfoListSizeExt* in *SIB12*), if *SIB12* is available.

For inter-UE coordination (IUC) information configuration, the UE may obtain it from *SIB12* (for in-coverage UE, as defined in clause 8.2, in RRC_IDLE and RRC_INACTIVE state; or for out-of-coverage UE, as defined in clause 8.2, on the frequency which UE is configured to perform NR sidelink communication and which is included in *sl-FreqInfoList*/*sl-FreqInfoListSizeExt* in *SIB12*) or *SL-PreconfigurationNR* (for out-of-coverage UE, as defined in clause 8.2, on the frequency which UE is configured to perform NR sidelink communication and which is not included in *sl-FreqInfoList*/*sl-FreqInfoListSizeExt* in *SIB12*).

For ranging/sidelink positioning, the UE may obtain the configuration from *SIB12* or *SIB23* (for in-coverage UE, as defined in clause 8.2, in RRC_IDLE and RRC_INACTIVE state) or *SL-PreconfigurationNR* (for out-of-coverage UE, as defined in clause 8.2).

8.2 Cell selection and reselection for Sidelink

The requirements defined in this clause for sidelink operation (including sidelink relay operations) apply for UEs in RRC_IDLE, RRC_INACTIVE and in RRC_CONNECTED.

When UE is interested to perform NR sidelink communication/discovery and ranging/sidelink positioning on non-serving frequency, it may perform measurements on that frequency or the frequencies which can provide inter carrier NR sidelink configuration for that frequency for cell selection and reselection purpose in accordance with TS 38.133[8]. When UE is interested to perform V2X sidelink communication on non-serving frequency, it may perform measurements on that frequency or the frequencies which can provide inter carrier V2X sidelink configuration for that frequency for cell selection and intra-frequency reselection purpose in accordance with TS 38.133[8].

If the UE detects at least one cell on the frequency which UE is configured to perform NR sidelink communication/discovery and ranging/sidelink positioning on fulfilling the S criterion in accordance with clause 8.2.1, it shall consider itself to be in-coverage for NR sidelink communication/discovery and ranging/sidelink positioning on that frequency. If the UE cannot detect any cell on that frequency meeting the S criterion, it shall consider itself to be out-of-coverage for NR sidelink communication/discovery and ranging/sidelink positioning on that frequency.

If the UE detects at least one cell on the frequency which UE is configured to perform V2X sidelink communication on fulfilling the S criterion in accordance with clause 8.2.1, it shall consider itself to be in-coverage for V2X sidelink communication on that frequency. If the UE cannot detect any cell on that frequency meeting the S criterion, it shall consider itself to be out-of-coverage for V2X sidelink communication on that frequency.

If the UE has selected a cell on a non-serving frequency for V2X sidelink communication, it shall perform additional intra-frequency reselection process to select a better cell for sidelink operation on that frequency in accordance with clause 8.2.1.

If the UE has selected a cell on a non-serving frequency for NR sidelink communication/discovery, it shall perform additional reselection process to select a better cell for sidelink operation in accordance with clause 8.2.1.

If the UE has selected a cell on a non-serving frequency for Ranging/Sidelink Positioning, it shall perform additional reselection process to select a better cell for sidelink operation in accordance with clause 8.2.1.

8.2.1 Parameters used for cell selection and reselection triggered for sidelink

When evaluating S criterion, R criterion (ranking) or inter-frequency cell reselection criterion, as defined in clause 5.2.3.2, clause 5.2.4.6 and clause 5.2.4.5 respectively, for cell selection/reselection triggered for NR sidelink communication/discovery or V2X sidelink communication or Ranging/Sidelink positioning on a non-serving frequency, UE shall perform the evaluation as follows:

- The UE shall use cell selection/reselection parameters broadcast by the concerned cell (i.e. selected cell for the sidelink operation) for the evaluation.

9 Tracking Reference Signal

The UE in RRC_IDLE and RRC_INACTIVE states may use Tracking Reference Signal (TRS) whose configurations are provided in system information for its paging reception to save power. In a cell in which TRS are available for the UE in RRC_IDLE and RRC_INACTIVE states to use, the availability of the configured TRS is informed to the UEs in RRC_IDLE and RRC_INACTIVE states based on explicit L1 based availability indication defined in TS 38.213 [4].

Annex A (informative): Example of Hashed ID Calculation using 32-bit FCS

Inputs:

- Least significant bits of 5G-S-TMSI: 0x12341234
- Generator polynomial: 0x104C11DB7 (1 0000 0100 1100 0001 0001 1101 1011 0111)

Procedure to Calculate Hashed ID:

step a)

- k = 32
- numerator: 0xFFFF FFFF 0000 0000
- denominator: 0x1 04C1 1DB7
- remainder Y1 = 0xC704DD7B

step b)

- numerator: 0x1234 1234 0000 0000
- denominator: 0x1 04C1 1DB7
- remainder Y2 = 0x1D66F1A6

Hashed_ID = FCS = ones complement of (remainder Y1 XOR remainder Y2)

= ones complement of (0xC704DD7B XOR 0x1D66F1A6)

= negation of (0xDA622CDD)

= 0x259DD322

Annex B (informative): Change history

Change history							
Date	Meeting	TDoc	CR	Rev	Cat	Subject/Comment	New version
3/2017						Initial skeleton	0.0.1
5/2017						Updated initial skeleton	0.0.2
6/2017						Updated based on RAN2#98 agreements	0.0.3
8/2017						Updated based on feedback from companies	0.0.4
10/2017						No changes	0.0.5
11/2017						No changes	0.0.6
01/2018						No changes	0.0.7
01/2018						No changes	0.1.0
02/2018						Updated based on RAN-NR-AH1801 agreements	0.1.1
02/2018						No changes	0.1.2
03/2018						Updated based on RAN#101 agreements	0.2.0
03/2018	RAN#79	RP-180451				Typo corrections; submitted to RAN#79 for information	1.0.0
04/2018	RAN2#101bis	R2-1805086				No changes	1.0.1
05/2018	RAN2#102	R2-1806884				Updated based on RAN2#101bis agreements	1.1.0
06/2018		R2-1809262				Updated based on RAN2#102 agreements	1.2.0
06/2018	RP#80	RP-180694				Submitted to RAN#80 for approval	2.0.0
06/2018						Upgraded to Rel-15 after the plenary approval	15.0.0
09/2018	RP-81	RP-181941	0024	2	F	Miscellaneous Corrections based on endorsed CRs in RAN2#103	15.1.0
12/2018	RP-82	R2-1818509	0047	4	F	Clarification of Paging Monitoring Occasion	15.2.0
	RP-82	R2-1816301	0049	2	F	Correction to description of PO for default association	15.2.0
	RP-82	R2-1819196	0051	3	F	Corrections on 38.304	15.2.0
	RP-82	R2-1816678	0055	2	F	CR on PDCCH monitoring occasions for paging	15.2.0
	RP-82	R2-1817200	0056	2	F	Release and Redirect in 2-step procedure	15.2.0
	RP-82	R2-1818681	0062	4	F	Clarification on final suitability check	15.2.0
	RP-82	R2-1817261	0063	2	D	Correction to Ambiguous Terminologies with respect to Cell Ranking	15.2.0
	RP-82	R2-1818125	0066	2	F	Correction on definition of PEMAX1, PEMAX2	15.2.0
	RP-82	R2-1817662	0067	2	F	Clarification of mobility state detection criteria	15.2.0
	RP-82	R2-1818549	0074	2	F	CR on UE behaviour upon lack of TAC in SIB1	15.2.0
	RP-82	R2-1818508	0075	1	F	Miscellaneous Corrections in Paging	15.2.0
	RP-82	R2-1819037	0076	1	F	Clarification for the UE behaviour in camped normally and camped on any cell states	15.2.0
	RP-82	R2-1818883	0079	2	F	Corrections on storing and discarding UE AS context	15.2.0
	RP-82	R2-1818998	0084	2	F	Correction to UE behavior for barred cell	15.2.0
	RP-82	R2-1817141	0087	-	F	Clarification on cell reselection conditions during TreselectionRAT	15.2.0
	RP-82	R2-1818881	0088	2	F	Introduction of MICO mode	15.2.0
	RP-82	R2-1817145	0089	-	F	Correction to monitoring paging in Camped on Any Cell state	15.2.0
	RP-82	R2-1818878	0090	1	F	Minor clarifications for paging	15.2.0
	RP-82	R2-1818688	0096	1	F	Correction to cell access barring alleviation in cell reselection	15.2.0
	RP-82	R2-1818683	0097	1	F	CR on 38.304 corrections	15.2.0
	RP-82	R2-1817738	0099	-	F	38.304 CR on derivation of N for paging	15.2.0
	RP-82	R2-1819026	0102	1	F	Correction of PLMN selection in RRC_INACTIVE	15.2.0
03/2019	RP-83	RP-190540	0081	4	F	CR on UE behavior if emergency call is not supported in the current cell	15.3.0
	RP-83	RP-190540	0103	2	F	Correction to number of radio frames spanned by PDCCH monitoring occasions of a PO	15.3.0
	RP-83	RP-190540	0104	1	F	Correction to signaling aspects of parameter first-PDCCH-MonitoringOccasionOfPO	15.3.0
	RP-83	RP-190540	0111	-	F	Clarification for paging and UL symbols	15.3.0
	RP-83	RP-190544	0113	3	F	Miscellaneous Corrections	15.3.0
	RP-83	RP-190544	0117	1	F	Correction on exception to cell categories for regional provision of service	15.3.0
06/2019	RP-84	RP-191373	0120	-	F	UE behavior on cell reselection if rangeToBestCell is configured_Option 2	15.4.0
	RP-84	RP-191373	0121	-	F	Miscellaneous Corrections	15.4.0
	RP-84	RP-191374	0125	-	F	CR on inter-RAT cell reselection	15.4.0
	RP-84	RP-191376	0126	1	F	UE behaviour on the cell without TAC	15.4.0
	RP-84	RP-191376	0128	1	F	Correction for Access Identity 0 when PLMN is reserved for operator use	15.4.0
	RP-84	RP-191376	0132	1	F	Monitoring of short messages with multi-beams	15.4.0
09/2019	RP-85	RP-192193	0136	2	F	Miscellaneous Corrections	15.5.0
12/2019	RP-86	RP-192936	0139	2	F	Miscellaneous Corrections	15.6.0
	RP-86	RP-192938	0143	1	F	Correction to Pcompensation for FR2	15.6.0
03/2020	RP-87	RP-200344	0145	4	B	Introduction of UE Power Saving in NR	16.0.0
	RP-87	RP-200353	0148	2	B	Introduction of PRN for TS 38.304	16.0.0
	RP-87	RP-200341	0149	2	B	Introduction of NR operation with Shared Spectrum Access in Idle/Inactive mode	16.0.0
	RP-87	RP-200349	0150	3	B	Correction of TS 38.304 to introduce IAB	16.0.0

	RP-87	RP-200346	0151	1	B	CR on cell selection/ reselection for NR V2X UE	16.0.0
07/2020	RP-88	RP-201179	0153	4	F	Miscellaneous correction to 38.304 for IAB	16.1.0
	RP-88	RP-201165	0155	4	A	Corrections to cell barred handling	16.1.0
	RP-88	RP-201182	0156	3	F	Corrrection to 38.304 for PRN	16.1.0
	RP-88	RP-201179	0157	4	B	Corrections to 38.304 for supporting IAB in NPN	16.1.0
	RP-88	RP-201174	0158	1	B	CR for UE Power Saving in NR	16.1.0
	RP-88	RP-201190	0173	-	C	Introduction of eCall over IMS for NR	16.1.0
	RP-88	RP-201176	0174	2	F	CR on cell (re)selection for sidelink in TS 38.304	16.1.0
09/2020	RP-89	RP-201927	0175	1	F	Correction to TS 38.304	16.2.0
	RP-89	RP-201932	0184	1	D	Miscellaneous corrections (Rapporteur)	16.2.0
	RP-89	RP-201923	0185	-	F	Miscellaneous corrections for TS 38.304 for IAB	16.2.0
	RP-89	RP-201929	0186	1	F	CR for UE Power Saving in NR	16.2.0
	RP-89	RP-201928	0187	-	F	Idle mode corrections for NPN	16.2.0
12/2020	RP-90	RP-202776	0193	1	F	Correction on RRM relaxation	16.3.0
	RP-90	RP-202771	0195	-	F	Miscellaneous Corrections	16.3.0
	RP-90	RP-202769	0196	-	F	Correction on inter-frequency operation	16.3.0
03/2021	RP-91	RP-210689	0203	-	F	Correction to 38.304 on intra-frequency reselection	16.4.0
06/2021	RP-92	RP-211471	0206	1	F	CR on the missing definition of "Available SNPN" in TS 38.304	16.5.0
	RP-92	RP-211475	0211	2	F	Correction of IFRI-related conditions	16.5.0
09/2021	RP-93	RP-212442	0215	1	F	Clarification of access restrictions during cell re-selection	16.6.0
	RP-93	RP-212439	0217	1	A	Clarification of barring when TAC is missing in RAN sharing	16.6.0
	RP-93	RP-212438	0220	-	A	Correction to cell selection and reselection due to SIB1 acquisition failure	16.6.0
12/2021	RP-94	RP-213343	0222	-	F	Correction for TS 38.304 on power class for cell selection of IAB	16.7.0
03/2022	RP-95	RP-220506	0204	2	D	Inclusive Language Review for TS 38.304	17.0.0
	RP-95	RP-220484	0221	7	B	Introduction of NR MBS into 38.304	17.0.0
	RP-95	RP-220837	0223	1	B	Introduction of mobility-state-based cell reselection for NR HSDN [NR_HSDN]	17.0.0
	RP-95	RP-220837	0226	1	B	Introduction of MINT [MINT]	17.0.0
	RP-95	RP-220483	0227	1	B	Introduction of ePowSav in TS 38.304	17.0.0
	RP-95	RP-220472	0228	1	F	Correction on PO determination for UE in inactive state	17.0.0
	RP-95	RP-220493	0230	1	B	Introduction of Enhancements for Private Networks	17.0.0
	RP-95	RP-220491	0232	1	B	38.304 CR for SL Relay	17.0.0
	RP-95	RP-220482	0233	2	B	Introduction of NTN	17.0.0
	RP-95	RP-220480	0234	1	B	Introduction of RedCap	17.0.0
	RP-95	RP-220490	0235	1	B	Introduction of slice-based cell re-selection	17.0.0
	RP-95	RP-220476	0236	-	B	Introduction of NR Sidelink enhancements	17.0.0
06/2022	RP-96	RP-221727	0238	1	F	Corrections on TS 38.304 for ePowSav	17.1.0
	RP-96	RP-221719	0246	4	C	38.304 CR Corrections on slice-based cell reselection	17.1.0
	RP-96	RP-221727	0248	1	A	Addressing inconsistency for RRM measurement rules	17.1.0
	RP-96	RP-221736	0249	1	D	Miscellaneous Editorial Corrections	17.1.0
	RP-96	RP-221729	0251	2	F	Alignment of DRX for Paging with RRC for SDT	17.1.0
	RP-96	RP-221716	0252	2	F	Miscellaneous corrections for RedCap WI	17.1.0
	RP-96	RP-221717	0254	-	F	NTN corrections to 38.304	17.1.0
	RP-96	RP-221732	0255	-	F	Miscellaneous correction on SL relay	17.1.0
09/2022	RP-97	RP-222523	0256	2	F	38.304 Corrections for MBS	17.2.0
	RP-97	RP-222525	0267	1	F	Miscellaneous CR on TS 38.304 for ePowSav	17.2.0
	RP-97	RP-222525	0276	1	F	Miscellaneous corrections for RedCap WI	17.2.0
	RP-97	RP-222524	0277	1	F	Miscellaneous corrections on 38.304	17.2.0
	RP-97	RP-222525	0280	2	F	Cell reselection corrections to RAN slicing	17.2.0
12/2022	RP-98	RP-223412	0288	3	F	Corrections on 38.304 for SL relay	17.3.0
	RP-98	RP-223411	0296	2	F	Corrections to TS 38.304 for NR NTN	17.3.0
	RP-98	RP-223410	0297	1	F	MBS corrections for 38.304	17.3.0
	RP-98	RP-223413	0298	1	D	Miscellaneous CR on TS 38.304 for ePowSav	17.3.0
	RP-98	RP-223411	0301	1	F	Correction on iPo determination for UE operates with eDRX	17.3.0
	RP-98	RP-223412	0302	1	F	Clarification on NSAG information in slice-based cell reselection	17.3.0
	RP-98	RP-223406	0310	1	F	Clarification on cell reselection priority handling for HSDN, MBS, V2X/NR sidelink, Slicing and deprioritization request	17.3.0
	RP-98	RP-223411	0313	1	F	Miscellaneous corrections for RedCap WI	17.3.0
03/2023	RP-99	RP-230690	0318	2	F	CR to 38.304 on relaxed measurements	17.4.0
	RP-99	RP-230691	0323	1	F	Correction on eDRX	17.4.0
	RP-99	RP-230692	0324	2	F	Correction to slice-support cell lists	17.4.0
06/2023	RP-100	RP-231415	0330	1	F	Addition of slice-based cell re-selection parameters	17.5.0
	RP-100	RP-231416	0333	3	F	Clarification on sidelink communication resource configuration used by OoC L2 Remote UE	17.5.0
	RP-100	RP-231417	0334	2	F	Corrections for eDRX in RRC_INACTIVE	17.5.0
	RP-100	RP-231410	0341	-	A	Correction on TS 38.304 for NR SL	17.5.0
	RP-100	RP-231411	0343	2	A	Clarification on Access Identities Validity	17.5.0
	RP-100	RP-231416	0346	-	F	Corrections on TS 38.304 for SL enhancements	17.5.0
	RP-100	RP-231415	0348	-	F	Clarifications on the use of SIB16	17.5.0
09/2023	RP-101	RP-232570	0351	2	F	NSAG validity when TAI list is omitted	17.6.0
12/2023	RP-102	RP-233888	0353	3	F	Correction on SIB/Preconfiguration applicability	17.7.0

	RP-102	RP-233888	0363	1	F	Clarification for the use of term and/or within the context of (e)DRX operation	17.7.0
12/2023	RP-102	RP-233898	0345	3	B	Introduction of NCR in TS 38.304	18.0.0
	RP-102	RP-233907	0355	1	**B**	Introduction of eMBS	18.0.0
	RP-102	RP-233912	0356	-	**B**	Introduction of R18 eNPN for TS 38.304	18.0.0
	RP-102	RP-233906	0357	2	**B**	Introduction of NR NTN enhancements in 38.304	18.0.0
	RP-102	RP-233901	0358	1	**B**	Introduction of R18 positioning to RRC_IDLE mode and RRC inactive state	18.0.0
	RP-102	RP-233897	0359	1	**B**	Introduction of Release-18 SL Evolution in TS 38.304	18.0.0
	RP-102	RP-233899	0364	2	**B**	Introduction of eRedCap in TS 38.304	18.0.0
	RP-102	RP-233904	0365	1	**B**	Introduction of Rel-18 support for SL Relay Enhancements	18.0.0
	RP-102	RP-233900	0369	1	**B**	Introduction of Network Energy Savings for NR	18.0.0
	RP-102	RP-233933	0370	-	**B**	Introduction of mobile IAB for NR	18.0.0
	RP-102	RP-233916	0371	-	**B**	Introduction of NR ATG in TS 38.304	18.0.0
03/2024	RP-103	RP-240731	0372	4	**F**	Restriction of cell list for NCR-MT cell reselection	18.1.0
	RP-103	RP-240695	0376	1	**F**	Miscellaneous Corrections for NTN in 38.304	18.1.0
	RP-103	RP-240657	0378	1	**A**	Clarification on the case SL frequency is not included in SIB12	18.1.0
	RP-103	RP-240700	0379	-	**D**	Correction on 38.304 for SL Relays	18.1.0
	RP-103	RP-240707	0382	4	**B**	Introduction of 2Rx XR UEs [2Rx_XR_Device]	18.1.0
	RP-103	RP-240657	0385	1	**A**	Correction on pre-configuration usage	18.1.0
	RP-103	RP-240662	0390	-	**F**	Corrections for Network Energy Savings in 38.304	18.1.0
	RP-103	RP-240696	0391	-	**F**	Correction to IDLE mode procedure for R18 positioning	18.1.0
	RP-103	RP-240655	0392	1	**A**	Minor correction for NTN in 38.304	18.1.0
06/2024	RP-104	RP-241543	0380	5	**B**	Introduction of barring exemption for RedCap UEs for emergency calls [RedCap_EM_Call]	18.2.0
	RP-104	RP-241543	0381	6	**B**	Introduction of barring exemption for (e)RedCap and 2RX XR UEs for emergency calls [EM_Call_Exemption]	18.2.0
	RP-104	RP-241577	0398	1	**F**	Mismatch of terminology between 38.304 and 38.331	18.2.0
	RP-104	RP-241544	0399	3	**B**	MBS operation with eDRX MICO	18.2.0
	RP-104	RP-241564	0401	-	**F**	Miscellaneous corrections on TS 38.304 for eRedCap	18.2.0
	RP-104	RP-241558	0403	2	**F**	Terminology alignment in 38.304 for NR-NTN	18.2.0
	RP-104	RP-241553	0405	-	**A**	Reference for User Service Description	18.2.0
	RP-104	RP-241571	0407	-	**F**	Miscellaneous Corrections (Rapporteur)	18.2.0
	RP-104	RP-241543	0408	-	**F**	Correction on cell status for 2Rx XR UE [2Rx_XR_Device]	18.2.0
09/2024	RP-105	RP-242240	0409	2	**F**	Correction to barring exemption for (e)Redcap and XR 2RX UEs [EM_Call_Exemption]	18.3.0
	RP-105	RP-242239	0411	-	**F**	Miscellaneous corrections on TS 38.304 for eRedCap	18.3.0
12/2024	RP-106	RP-243223	0414	6	**A**	Correction to Relaxed measurement	18.4.0
	RP-106	RP-243230	0421	1	**F**	Clarifications on cell barring behaviour [2Rx_XR_Device] [EM_Call_Exemption]	18.4.0
	RP-106	RP-243231	0422	2	**F**	Rapporteur CR to IDLE mode procedure for R18 Positioning	18.4.0
09/2025	RP-109	RP-252798	0439	1	**B**	Introduction of UAV mobility enhancements [UAV_Mobility]	19.0.0
	RP-109	RP-252773	0440	3	**B**	Introduction of LP-WUS in TS 38.304	19.0.0
	RP-109	RP-252778	0441	1	**B**	Introduction of Rel-19 NR NTN	19.0.0
	RP-109	RP-252774	0442	2	**B**	Introduction of Rel-19 network energy saving to TS 38.304	19.0.0
	RP-109	RP-252780	0444	1	**B**	Introduction of multi-hop U2N relay in TS 38.304	19.0.0
	RP-109	RP-252868	0446	1	**B**	Restriction on RAT utilization	19.0.0
12/2025	RP-110	RP-253729	0448	1	**F**	Correction on broadcast service in NTN	19.1.0
	RP-110	RP-253725	0450	1	**F**	Corrections on LP-WUS in TS 38.304	19.1.0
	RP-110	RP-253719	0451	1	**A**	Correction on application of restrictions to mIAB-MT	19.1.0